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College of Electrical & Mechanical Engineering
Department of CSIT, Software Engineering

Undergraduate Project Report

Title: Electronic Citizen Identification System (ECIDS)

Group Members:

No	Name	ID
1	Hadera Teame	ETS0471/09
2	Fithamlak Fikre	ETS0429/09
3	G/slasie Abreha	ETS0451/09
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Head, Department of Software Engineering

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Examiner

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Abstract

This paper is a documentation for Electronic Citizen Identification System (ECIDS). ECID is a system for maintaining citizen information. The main purpose of ECIDS is to facilitate the citizen ID process and solve the problem that arose using the former and current ID systems. This can be achieved by registering citizens, maintaining their information in efficient and organized form and then make citizen information available to authorized users indeed. This document contain purpose and objective, functional and non-functional requirements and design of ECIDS in different sections. Section one describes the general objective and purpose of the system, scope and limitation of the system and methodologies and tools used in the system. Section two describes the functional, non-functional requirement and constraint of the system. Section three describe different models that represent different features and functionalities of the system. Section four describes components and modules of the system.

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ይህ ሰነድ ስለ ኮምፑተራይዝድ የዜጎች መታወቂያ ሲይስተም የሚያትት ነው። ኮምፑተራይዝድ የዜጎች መታወቂያ ሲይስተም የዜጎች መረጃ በተቀናጀ እና በተደራጀ መልኩ ያስቀምጣል።የዚህ ሲይስተም ዋና ኣላማ የ መታወቂያ ኣሰጣጥ ሂደት ማቀላጠፍ እና በ ድሮው እና በ ኣሁኑ የመታወቂያ ስርዓቶች የሚታዩ ችግሮች መፍታት ነው። ይህ ኣላማ ዜጎችን በተቀላጠፈ መልኩ በ መመዝገብ ፣ የዜጎች መረጃ በተቀናጀ መልኩ በማስቀመጥ እና የዜጎች መረጃ ለ ተፈቀደላቸው ተጠቃሚዎች በተገቢው መልክ በ መስጠት ማሳካት ይቻላል። ይሄ ሰነድ የ ሲይስተሙ ግብ ወይም ኣላማ ፣ “ረኳየርመንት“ እና ዲዛይን በ ተለያዩ ክፍሎች ያካቲታል። ክፍል ኣንድ የሲይስተሙ ኣላማ እና ፋይዳ ይገልፃል።ክፍል ሁለት የሲይስተሙ መገለጫዎች እና ተግባራት ወይም በሌላ አገላለፅ “ረኳየርመንት“ ይተነትናል።ክፍል ሶስት የሲይስተሙ ቅርፅ ፣ መገለጫዎች እና ተግባራት የተለያዩ ናሙና ስእሎችን ወይም “ሞደል” በመጠቀም ያሳያል።ክፍል ኣራት የሲይስተሙ ክፍሎች ወይም ኣካላት በተብራራ መልክ ይገልፃል።

Acknowledgement

First and foremost, we would like to thank God Almighty for giving us strength, knowledge ability and opportunity to work and complete our project.

Next would like to express our special appreciation and thanks to our adviser Mrs. Yaynishet Medhin for his great support and advice to make our project. He helped us by granting advice and direction, correcting errors and generally guiding the project.

We also thank all people who help us working our project. We would like to thank citizens, government employee and authorities, police officers, organization, Lion bank's employee, commercial bank of Ethiopian's employee and Ethiopian airline for their willingness to share their idea, fill questionnaires and answer our questions.

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Acronym and abbreviation

Table 1 : Acronym and abbreviation

Acronyms and Abbreviations	*Stand for*
ID	Identification
MVC	Model View Controller
SRS	System Requirement Specifications
UML	Unified Modeling language
EA	Enterprise architecture
IDCO	Identification card office
ECIDS	Electronic citizen identification system
HTML	Hypertext markup language
API	Application programming interface
PC	Personal computer
HTTP	Hypertext Transfer Protocol
CSS	Cascading styling sheet
RQ	Requirement
US	Use case
ADMIN	Administrator

CHAPTER ONE: Introduction

1.1 Background

Uniquely identifying citizenship of citizens in any country is very important to facilitate various activities of the governmental, non-governmental organizations like Public Administration Services, Social Security Services, Medical Services, Tax Collection Services, Vital Registration Services, and much more. It cans also enable government agencies to provide these services with better integrity and control. Currently, developing such capabilities is possible because of the information technology revolution that allows sharing of large amount of information among computer networks. [1]

Identifying citizens is a key theme of twenty-first century life. Indeed, new identification system including national ID cards may turn out to be the single most significant development of information systems for governance, globally. Of course, the very advent of modern forms of citizenship assumes identification (Abercrombie et al. 1986, Caplan and Torpey 2001, Jenkins 2004). In modern times, bureaucratic administration demands that each person be identified to ensure that they are entitled to call themselves citizens. Without identification one can neither take on responsibilities, such as voting, nor enjoy the benefits, such as protection from external or internal threats to well-being. However, in the twenty-first century identification processes are changing shape in radical ways that take them far beyond what might have been experienced even towards the end of the twentieth century. [1]

Governments issuing Citizen Identification card can use it for different purposes such as efficient public transactions and border control. The concept of a citizen identification system was first instituted in countries with populations coming from diverse ethnic groups. Earlier, Group Classification was one reason for using ID systems. [2]

Electronic identification is a digital solution for proof of identity of citizens or organizations, for example in view to access benefits or services provided by government authorities, banks or other companies, for mobile payments, etc. Apart from online authentication and login, many electronic identity services also give users the option to sign electronic documents with a digital signature [3].

Countries which currently issue government-issued electronic identification include Afghanistan, Malta, Bangladesh, Belgium, Bulgaria, Chile, Finland, Guatemala, Germany, Indonesia, Israel, Italy, Nigeria, Luxembourg, the Netherlands, Mexico, Morocco, Pakistan, Portugal, Poland, Romania, Estonia, Latvia, Lithuania, Spain, Slovakia, and Mauritius. Germany, Uruguay and previously Finland have accepted government issued physical EIC: s. Norway, Sweden and Finland accept bank-issued Bank ID for identification by government authorities. There are also an increasing number of countries applying electronic identification for voting (enrollment, issuing voter ID cards, voter identification and authentication, etc.), including those countries using biometric voter registration. [3]

Currently in USA, the social security number (SSN) is serving as unique citizen identification number in most governmental and private institutions databases including Internal Revenue Service, Veteran Administrations, Hospitals, Banks, Insurances, etc. SSN is a nine-digit number resembling "123-00-1234" which is issued to an individual by the Social Security Administration of the government of the United States. Formerly, some people did not have SSN until they reach 15 or 16 years of age, as it was used for tax purposes and those people under 15 hardly had income generating employment. At present, many parents apply for social security numbers for their children as soon as the children are born. However, SSN was never intended to be a National ID number, although it is usually used for identification [4].

In Ethiopia, Addis Ababa formerly the identification system was manual and paper based and the system was not supported by computer. All citizen data were stored and organized in paper. Data stored in paper is not safe; it can be lost and damaged easily at the time of usage, transportation or storage. In this system searching and querying data from file was manual. This identification system was then modified.

Later the identification system was supported by computer software that can store and organize data at woreda level but it was still manual. The software was controlled and managed by individual employees who are specifically assigned to operate the software. This means that the software is dependent on the individuals who are able to operate it and if the individuals are absent the software do not provide service. In this identification system, various woreda and sub cities communicates and works together hardly because the software do not provide interface or a way to communicate with other woreda. Although this identification system slightly modify the manual system it was venerable to fake and illegal ID cards.

Currently in Ethiopia Addis Ababa, installation of digital identification system in progress. This digital identification system help woreda and sub cities to have strong communication. This system is modernized and solve many problems of old manual and semi-manual systems but still it have some drawbacks. The efforts made to introduce an Electronic ID System is not satisfactory Although some efforts are under study in to introduce Digital Identification system in some areas mainly in Addis Ababa and some of them are implemented .Some effort is made on the Citizen Identification System, which can greatly assist e-governance by availing a means to integrate citizens' information from different organization to facilitate the service rendering process of the government offices.

1.2 Statement of problem

Current Identification system is modern but still, it has drawback. It is time consuming, vulnerable to some illegal activities and citizen data are not available to be used by external user. The following paragraph explain the problems in detail.

Current identification system is time consuming. And most citizens prefers manual ID system to digital ID system because they do not want to spend many time. One person has to dedicate much hours, day even weeks to be registered and have the identification card. The registration system takes time. Because of the current system main intent is to avoid Identity theft and illegal actions; the registration process has to done carefully. Due to this there are a lot of processes and verification has to be passed in order to be registered. In current identification system prepared ID cards has to be signed and approved by higher official before they are given to citizen due to this it takes 1 up to 2 weeks to have new ID card. The registration, Id giving and other process takes time and waste citizen time.

Current Identification system has inefficient verification system. Citizens could make fake and illegal ID card and they could use them to get some service. In order to prevent this there should be some verification process to check whether a given card is valid or invalid. The current system uses some security code on ID cards and some notes on the ID card. But this method is inefficient since there are a lot of tricks to have fake IDs.

Current Identification system has no interface for external users and organizations. Data in current identification system are an accessible to external uses .Banks, aviation and other organizations use citizen data to provide service. But Identification system do not have interface they could not access citizen data and verify ID. Due to this many organization are exposed to many illegal actions and crimes.

Current Identification system organizes and stores limited amount information. There are a lot of citizen information that should be stored and ready for access indeed. Those information include Education status, crime status (crime record), health risk, blood type, marital status and others. Without those information it hard to full identity citizens. Current identification system maintains some information from the mentioned information, and it should maintain more information in order to identify citizens fully.

1.2.1 Existing system

The currently existing citizenship identification system registers the information of citizens in kebele level and organizes the data in database. This system changes the paper-oriented platform and no citizen can have more than one citizenship ID card, which prevents some problems, related to id fraud. On the other hand, this system does not support advanced verification and integration issues and has little efficiency.

Registration system with existing system

If a citizen is not registered on a file, he has to be register on new or existing file. There are criteria that should be fulfilled, before registering on new or existing file. One criterion is proving you reside in a given sub-city. Owing a house in that given sub-city and having house number could be a proof that you are a resident of that sub-city. So someone who has a house and resides in a given sub-city can be registered on a new file. Other criterion could be a report that shows that you lived in other sub-city of the country and you released from that sub-city and providing. But citizens could make and provide fake files, reports, certificates and documents. And then they could open file and have illegal ID card

Citizen identification process with existing system

Citizens, who can have new citizen ID cards, are those who already registered on their file or someone's file that is already created. So if a citizen needs to have new citizen ID card he goes to the organizations and searches for a file (the file could be his file or someone file where his name is registered). Then if the file is found, after some process he can take his ID card. But, since there is huge number of files it is hard to search for specific file. Even if the files are arranged in shelf in alphabetical order, it is not efficient because when time goes on the arranged files lose order incase searched and used files are not putted in the correct place.

Information of citizens with existing system

The existing system maintains basic citizen information like name, age, gender, job, address, birth place and other. In addition to this basic information it maintains citizen information like blood type. Any citizen who needs to register or who needs to have ID card has to come up with the medical report that shows about his blood type.

1.2.2 Major problems of the Existing system

The currently existing citizenship identification system has the following main problems

- **Lack of integration: -**Although it is starting using digital citizenship identification in some areas of Addis Ababa, it is not fully automated and integrated with governmental and non-governmental organizations like Banks, immigration agencies, hospitals and other institutions. Therefore, each organizations uses their own primary data or other keys as it stores. Thus, most of them still work in a manner similar to manual methods or semi-automation and independently.
- **Difficult to Authenticate: -** since the existing system doesn't support automatic checking techniques like barcode or related verification method, it is difficult to identify the authorized citizen, which means a certain citizen's card must be validated and searched manually in the database and this is very tedious and inconvenient.
- **Lack of Interface for external users or organization: -** the existing system organizes and stores citizen data, but it does not have interface for external users.
- **Sufficient citizen detail is not stored in database: -** the citizen data stored in database are limited. And important data like crime status and others**.**
- **Lack of modification: -** citizen's information change frequently and the changed information should be changed in the database. And one way or best time for updating citizen information is at the time of id renewal. The existing system

- **Consumes much time:** - major process like registration process, ID giving process and other process takes much time.

1.2.3 Proposed system

The proposed system will completely make the citizen identification system electronic. It will store and organize sufficient citizen detail in database, proof the citizenship of a person and give necessary information of citizens to external users in need.

Registration process with proposed system

The proposed system will issued to each citizen upon registration, as well as newborns, and including those who gain citizenship by registration or naturalization as well as those ordinary resident (person legally residing in Ethiopia or authorized to legally reside in Ethiopia for at least 6 month). Any person who can provide convincing document evidence can be registered by filling its personal details and uploading the associated document.

Citizen identification process with proposed system

Before anything happens, citizens should be registered on the system, after registration citizen are given unique registration identification number. Using that given identification number citizens can create account and get various services. Using that account citizen can request for ID card, see notification and get other service.

Citizen who are registered and who are above 18 can have ID card. They can request for ID card online and take their ID card when it is prepared. When citizen take ID card they are assigned to unique card identification number in addition to the registration identification number and also assigned unique bar code.

- The card identification number (card ID NO) identify uniquely those who have ID card
- The registration identification number (registration ID NO) identify uniquely those who are registered. Both identification numbers help as to identify citizens uniquely.

Institutions, organization as well as ordinary persons are able to check identity of one person with the proposed system simply by reading the barcode or entering card identification number.

The ID card (card identification number) of one's citizen is deactivated upon death or renunciation or other reason. But the identification number which uniquely identifies citizens will never be used again.

Information of citizens with proposed system

The information of each citizen will be registered on our system. Required information such as name, birth date, address, email, phone number, blood type, health risks and other additional information including photograph of the citizens are taken and stored.

Fetching data with the proposed system is swift enough which minimize the time wastage. In addition any data can be extracted simply using barcode reader instead of entering ID number manually. Institutions, organization as well as ordinary persons are granted to see information of citizens with some restriction. This implies that each of them are able to get information only that is concerned with them.

1.2.4 Advantage of proposed system

Electronic id is a secure, portable and inclusive document**.** The importance of the proposed system will arise from secure, portable and inclusive nature of the system.

- ID cards help associate a lot of information with a particular person. For Example, an ID card might carry information such as health risk that the individual holds [5].This information helps citizens during emergency time. The proposed system associate such information and also other additional relevant information with ID card.
- The proposed System will completely solve the issue of having two identities because upon registration each citizen is provided with unique registration identification number.
- Proposed system allows stored data to be accessed easily. By using efficient searching and querying method the yield system allows citizen's data that is stored in the database to be fetched easily when it is needed
- The proposed system minimize the time spent for getting service from identification system. By allowing citizens
    - ✓ Request for registration by filling information requirement and uploading document requirement. This reduce the time spent for registration since citizen fill all the data online instead of

- ✓ Request for service online. For example, citizen can request for ID card online.

1.3 Motivation

Observing people spent much time to get service, hearing many complaints from people, seeing institutions and organizations facing difficulties to get full information of citizens and generally suffering from problems arisen from the old and current identification systems, we believe that we should come up with best solution which solves all the mentioned problem.

Seeing all the advantage of electronic identification system we would like to bring these solutions to the everyday problem of our lives. By automating the current process and developing an electronic identification, we believe the entire process could only take up to a few minutes with higher efficiency and accuracy.

Currently government have intention to improve the identification system. So coming up with such modernized system is acceptable and profitable.

1.4 Scope and limitation of the project

1.4.1 Scope of the project

The system yield from this project is for

- Registering citizen as well as users and maintaining their data
- Providing away to access the maintained data.
- Providing environment to verify and validate ID cards.
- Allow citizen request for services online.
- Allow users upload their own personal information and document at time of registration.
- Generating and assigning unique ID numbers and barcodes

The system yield from this project is not for

- Performing document and information verification: - the system only uploads and stores information and document. Information and document verification is done by employees and admins.

- Getting full service online: - users can request for service online this minimize the service time. But the user has to go to office to get the full and actual service. For example use cannot be fully registered online. After it request for registration he has to go to office for approval.
- Updating personal information: - citizen and users do not modify their personal detail even their actual name (not user name). Modification is made by employees and admins.
- Preparing and printing ID cards: This system is not going to prepare, generate ID card or print Id card it is only concerned with providing necessary data for external system or API that prepares the manual card or paper and also maintaining or keep track of prepared identification cards.

1.4.2 Limitation of the project

The system that we are going to build has some limitations. The following are some limitations of our project

- Electronic identification cards have microchip, magnetic stripe and barcode. Our system is not going to use ID cards having barcode, magnetic stripe and microchip as most of the electronic ID system use due to some budget and economical limitation. Printing and preparing such ID card is expensive.
- The mobile application of this system is not build for current release of the system. The mobile application of the system is not included in this version because of time limitation. Mostly we are affected by the covid-19 pandemic and the country current stability problem also affected as. The mobile application are expected to be released on the coming version or release.
- Some requirements having low priority are not implemented on this version because of the time limitation other limitation we mentioned above.

1.5 Objective

1.5.1 General objective

Our project goals is to come up with a software solution to problems and difficulties appears working with current digital identifications system by building a new system that improve the current systems.

1.5.2 Specific objective

- Explore, identify and analyze problems of the old and current systems.
- Study and find out the major problems. And analyses the major reasons of those problems.
- Study advantage and possibility of implementing such electronic identification system.
- Provide web based application that help the employees to give service to its customers, help user login get some services.
- Provide mobile applications which is part of the system and part of the solution.

1.6 Methodology

1.6.1 Data collection methodology

By going to the target offices we observed all the activities and process made in those offices. By the observation we made, we abled to see how the current system works, what problem does it have and what difficulties faced by many stakeholders. The observation were taken at both Bole sub-city and Akaki-Kaliti sub-city.

We gather requirement using questionnaire. Four category of paper questionnaire were prepared to gather requirement from citizens, employees, external organization and police offices. From the survey we are able to gather possible requirements. Survey made both at target offices and outside target office. The prepared questionnaire paper are attached on Appendix 1.

We distributed questionnaire paper to

- Employees in Bole sub-city and Akaki-Kaliti sub-city

- Citizens inside target office(citizens come to get service) and citizens outside target office (any users or citizens)
- Bank (Ethiopian commercial bank and Lion international bank) and aviation (Ethiopian airlines)
- Police officer who were at work

We also made interview with higher official of Bole and Akaki-Kaliti sub-cities to get statistical data and to get clear understanding of the current system and former older system that were used by the sub-cities.

1.6.2 System design and analysis tools

- **Microsoft word:-** used for preparing system requirement specification , for writing proposal , for documenting
- **Enterprise Architecture: -** The UML diagram of our system was drawn with the help of enterprise architecture which is best for designing system and application in object oriented (OO) method.
- **Gantt chart: -** used for drawing project schedule and charts which is critical and important in project management.
- **Drawing tools: -** drawing tools such as paint used for drawing design pictures and preparing design prototype.

1.6.3 System development tools

- **React JS: -** Java script library used as to develop our web base application front end. We choose this frame because it make user interface responsive and attractive.
- **Visual Studio code: -** Text editor used for editing and running our code. This text editor has terminal or command promote integrated with it which makes our work easy.
- **GitHub: -** version control system which helps as manage changes made to our code.
- **Node JS: -** Java scrip frame work used for developing the back end of web based application system.
- **Mongo Db: -** No-SQL and non-relational database that minimize the time spent to query or fetch data.

CHAPTER TWO: System requirement specification

2.1 Background

The Electronic identification system is web based application and mobile application build to maintain citizen and user data. At the first place the system should register users and citizens. Citizens are main actors of this system whose data is maintained, organized and ready for access. Users are those people who are registered in order to access citizens' data, get some service or give service using the system. The system should maintain data of both citizen and users' data. Maintained data should be accessible. Access for data should be authorized and authenticated.

2.2 Functional requirement

- ***RQ 1:*** Register different users of the system
  - ***RQ 11:*** The system should register super Admin, so that super admin can add Admin and employee or do other works. There are two admin super admin and branch admin. System register should register both of them.
  - ***RQ 12:*** The system should register admin. Those are team leader who coordinates and control team of employees.
  - ***RQ 13:*** The system should register employees. Employees should be registered so that they can login and give various services. Employees request for registration by inserting their necessary details. Then super admin can confirm or approve their request and they can be registered.
  - ***RQ 14:*** System should register citizens. Citizens can be registered in two ways.
    - ✓ Employee can add citizen by inserting necessary information and uploading document requirement.
    - ✓ Citizen request for registration online and after verification if their request is accepted they can be registered.
  - ***RQ 15:*** System should register police officers which uses citizen information.
  - ***RQ 16:*** System should register organization which uses citizen information.
- ***RQ 2:*** Login and authentication. The system should allow the different user (people who have direct contact with the system) Login, authenticate and check their authority.

  - ***RQ 21:*** Login for Super Admin
  - ***RQ 22:*** Login for Admin
  - ***RQ 23:*** Login for Employee
  - ***RQ 24:*** Login for citizen
  - ***RQ 25-*** Login for organization
  - ***RQ 26-*** Login for police officer
- ***RQ 3:*** Generate unique ID number. The system should generate unique ID number strings using some parameters and assign it to users and ID cards.
- ***RQ 4:*** Prepare ID card. The system should prepare ID card for citizens. The system should organize and put the necessary information and data that should be displayed on the card and then provide the data to be printed to the external machine.
- ***RQ 5:*** Modify citizen information. Employees can modify citizen information if citizen can provide convincing reason for modification.
- ***RQ 6*****:** Customize user account. .For example users can change their user name and password.
  - ***RQ 61:*** Employees can customize their account.
  - ***RQ 62:*** Admin can customize their account.
  - ***RQ 63:*** Super Admin can customize its account.
  - ***RQ 64:*** Citizens can customize their account.
  - ***RQ 65:*** Organizations can customize their account.
  - ***RQ 66:*** Police officer can customize their account.
- ***RQ 7:*** Request for ID card online. Citizen can request for ID card online by inserting all necessary information. Then they can have their ID card by going to the IDCC if an employee approve their request by checking their eligibility.
- ***RQ 8:*** Provide citizen information to users. The system should provide or display citizen information to users who are registered and have permission. But the system should provide limited and only needed information and should keep citizen privacy.
  - ***RQ 81:*** Provide limited information to organizations.
  - ***RQ 82:*** Provide limited information to police officer.
  - ***RQ 83:*** Provide limited information to other citizens.
  - ***RQ 84:*** Provide full citizen information to employee.
  - ***RQ 85:*** Provide full citizen information to admin.

    - ***RQ 86:*** Provide full citizen information to super admin.
- ***RQ 9:*** Generate barcode. System should generate unique barcode that is going to be displayed on the citizen ID card and identify citizen uniquely.
- ***RQ 10:*** Validate citizen ID card. The system should check weather ID card is legal, fake, blocked and black listed. System should check whether ID card are valid and can be used.
- ***RQ 11:*** Block ID card and citizen. If there is some issues related to citizen and ID card must be out of use, system should ban or block the citizen and ID card. And also there should be away to unblock citizen and ID card if the issues is solved.
- ***RQ 12*:** Post notification. Admins can post different type of notifications.
    - ***RQ 121:*** Post public notifications.
    - ***RQ 122:*** Send notification to specific user or citizen.
- ***RQ 14:*** Display notifications. User can see new feeds and notifications.
- ***RQ 15:*** Renew citizen ID card. Citizen ID card should be renewed at interval of two year. System should provide a way to renew citizen ID cards. At the time of renewal every changed information of citizen in that interval will be updated.
- ***RQ 16:*** system set timer for citizen id of two years and set it as blocked and should be renewed so that the counter begins again.
- ***RQ 17:*** System should search citizens by their ID number, name and other attribute.
- ***RQ 18:*** Set citizen as wanted. If police is looking for suspect citizen, system should set citizens as wanted then anyone who see the citizen can report.
- ***RQ 19***: Set service ban. If there are issues related to a citizen and citizen should be banned from some services, system should ban the citizen from that service. For example if citizen has immigration ban, system should add information to database that the citizen has immigration ban. So that the citizen cannot be served by aviation and bus-stations.
- ***RQ 20***: Generate report. System should generate two type of report.
    - ***RQ 201:*** Generate transaction report.
    - ***RQ 202:*** Generate statistical report.

2.3 Non-Functional Requirements

- **Availability:-**The system and its data should be available all time and all place.

- **Security*:*** access to data should be controlled and only authorized user should access and modify data. The system should
  - ✓ Authentication user to filter out unauthorized user.
  - ✓ Record all activities and transaction, for analysis, recovery and tracking purpose.
  - ✓ Limit the data to be accessed. Client should get only data concerned with them.
- **Performance*:*** - The system must be able to run without interruption and able to handle thousands of concurrent requests of citizens. In addition the system should
  - ✓ Fully loads web pages in few seconds
  - ✓ Responses to queries in short time
- **User friendliness**: - System user interface should be interactive, attractive and responsive. User interface for all users should be different and should be convenient, understandable to each of them.
- **Maintainability:** The system should be designed in such a way that can be maintained easily by developers. The concept of modularity and component-based approach must be used in order to make the system easily maintainable and modifiable whenever there is a need for modification.
- **Mobility: -** The system should be easy to use it with mobile phones, tablets and personal computer. The mobile application of this system should not be platform dependent and should operate well in all mobile operating system.

2.4 Feasibility Study

Feasibility study for a system determines the strengths and weaknesses of proposed venture, opportunities, risk involved in running a project and assets needed to carry through a project [6]. The feasibility study of the proposed system determines whether a project is financially, technically, delivery time and operationally feasible under a set of assumption, such as the technology used, the facilities and equipment, the capital needs, and other financial aspects. Feasibility study of the proposed system can be analyzed with the following parameters.

Technical Feasibility

The technology for developing the proposed system is mature enough or practical, easily accessible and can be easily applicable. Proposed system can be accessed through smart phones, tablets, and computers hence users can access this system using their resources. In addition this system need few database server, application server, computers for officers and barcode reader. Thus, the proposed system is technically feasible.

Economic Feasibility

The proposed system cost not much amount of resource (time, money, effort). Ethiopian government can afford the cost of the proposed system. System has much profit since it minimize the cost of service time, the cost of effort (employee labor) and cost for illegal action and crimes. Thus proposed system is economically feasible.

Legal Feasibility

The proposed system considers the copyright restrictions, license agreements, and contracts. Proposed system requirements are specified on the basis of the current rules and laws. The system operation do not contradict with current law of the country. So the proposed system is legally feasible.

Schedule feasibility

Project schedule estimations in project proposal shows that, the proposed system can be fully developed and ready for deployment. Thus the proposed system is schedule feasible.

Behavioral feasibility

Proposed system is user-friendly. Ordinary users of proposed system are only expected to have basic understanding of computer not more than that. Support and tanning will be offered to technical staff (employees and admins). Thus proposed system is behavioral feasible.

Operational Feasibility

The proposed system is almost similarity with the current system except it modify and add some features. The current system is operating well. There is nothing that preventing the proposed system from operating well since similar system is in use. Thus proposed system is operational feasible.

CHAPTER THREE: System Analysis and Modeling

3.1 Overview

In this chapter the proposed will be modeled with various modeling techniques in order to design the system and represent or model the system requirement. Different model such as use-case diagram, activity diagram, sequence diagram, state diagram and class diagram are used to represent or model different aspects of the system.

3.2 Scenario Based Modeling

3.2.1 Use case Identification

- UC 1- Register
- UC 2 - Login
- UC 3 - Generate barcode
- UC 4 - Generate ID number
- UC 5 - Prepare ID card
- UC 6 - Modify
- UC 7 - Customize account
- UC 8 - Request for ID card
- UC 9 - Display
- UC 10 - Verify ID card
- UC 11 - Block ID card
- UC 12 - Post notification
- UC 13 - Renew ID card
- UC 14 - Search
- UC 15 - Set citizen as wanted
- UC 16 - Ban service
- UC 17 - View report

3.2.2 Actor Identification and Description

Employees: - These actors are who manly operates system and who initiate and perform many operation of the system. They have access to

- Add, delete, and modify citizen information.
- Add citizens and other users of the system.
- View full citizen information.

Police officers: -Those are any polices offices who have legal police ID card. They can ask citizen ID card at any time at any place. Taking that ID card they can

- Check the legality, validity and date of ID card.
- Request for banning citizen from services and request for making citizen wanted
- Check identity of citizen
- See citizen detail by reading the barcode in the ID card using their telephone or barcode reader

Organizations: - These are any organization external to the governmental offices. These could be banks, aviation, and others. These organizations ask citizen ID card in order to provide a service. Sometimes they may need more information beyond the information provided on the ID card. For example they may need extra information to check the identity of the client. Using client card they can see more details that is concerned with them. They can

- Check legality, validity and data of ID cards
- See from what services does the client banned. For example client could be banned from immigrating and aviation could check whether clients have or haven't immigration ban.

Citizen: - These are main actors of the system whose data is stored and maintained by the system. Citizens can be both

- Citizen: - as citizen their data is maintained in system database. And they get citizen service from the system.
- External user:-as external user thy can create account check others citizen identity and view limited amount of citizen detail.

Admin: - These are similar to ordinary employee except they are team leader and have higher access to some services and data than the ordinary employees. Viewing transaction and statically report, posting notification is exclusively given to them.

Super admin: - Secondary actor who administers the whole system operation and actor's activities. It has full access of monitor everything. Mainly he/she can

- Add, modify and remove employees and other system users
- See statistical report, transaction and activity report and user information

3.2.3 Use case diagrams

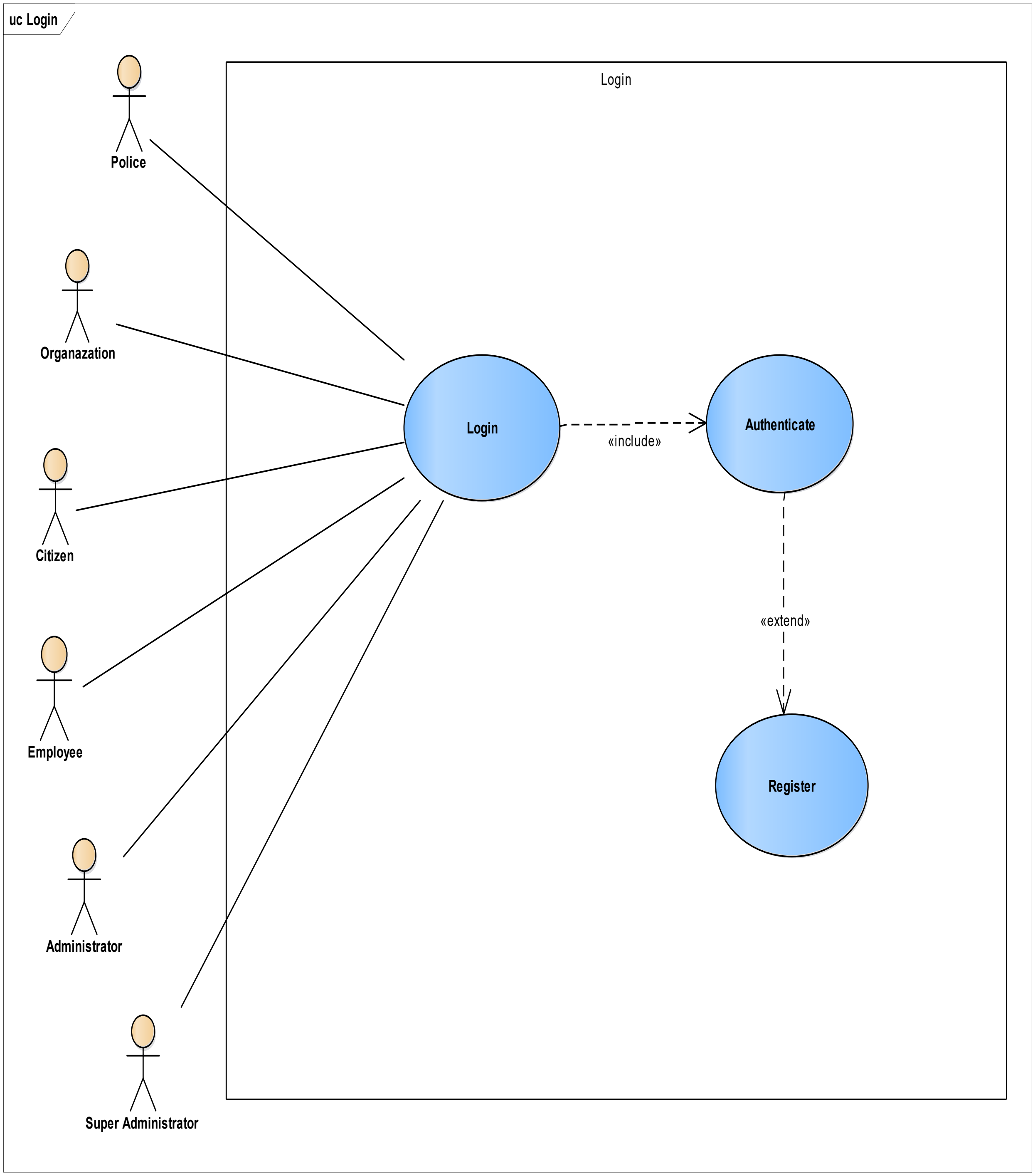

Figure 3. 1 Use case diagram for Login

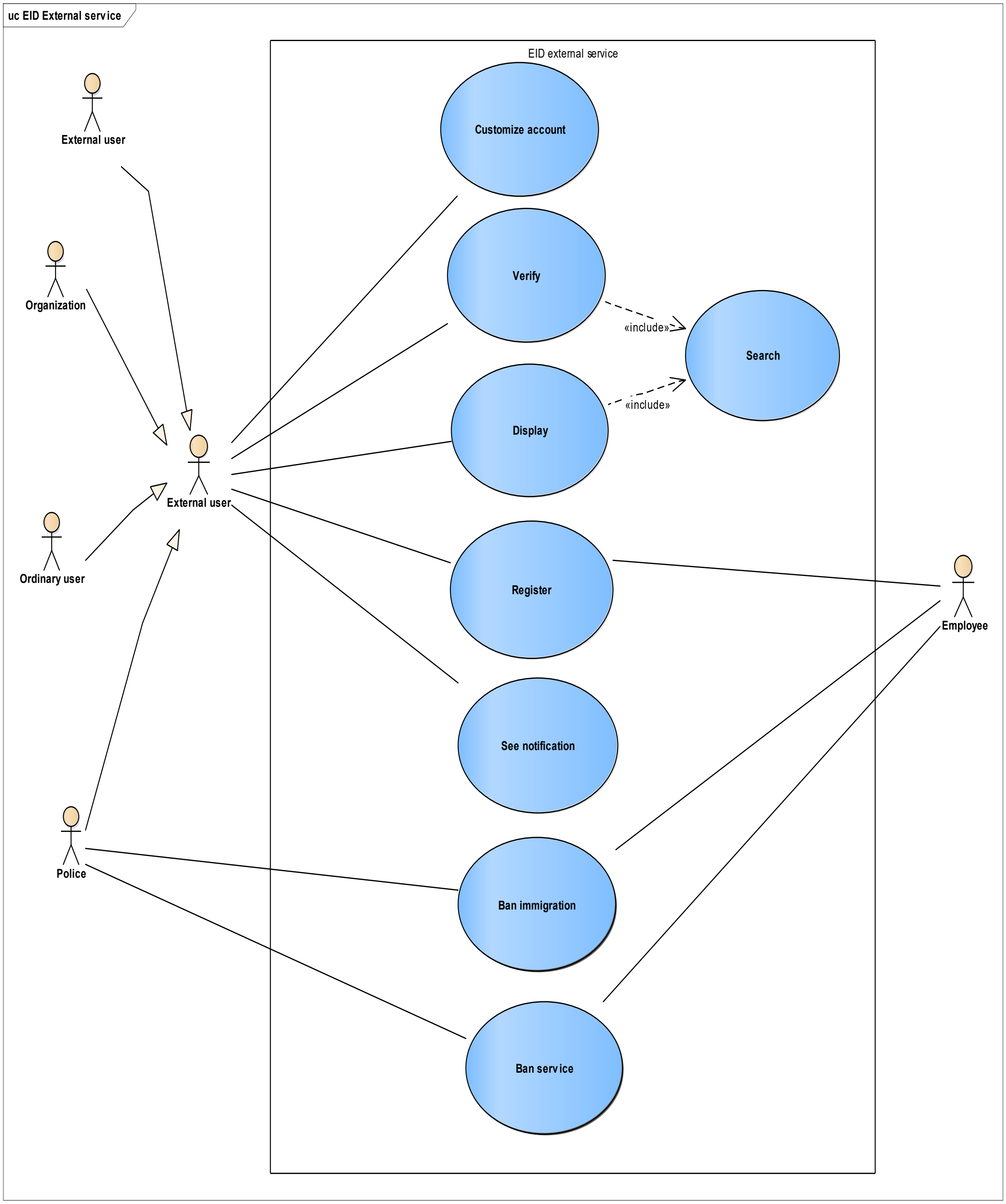

Figure 3. 2 Use case diagram for external service

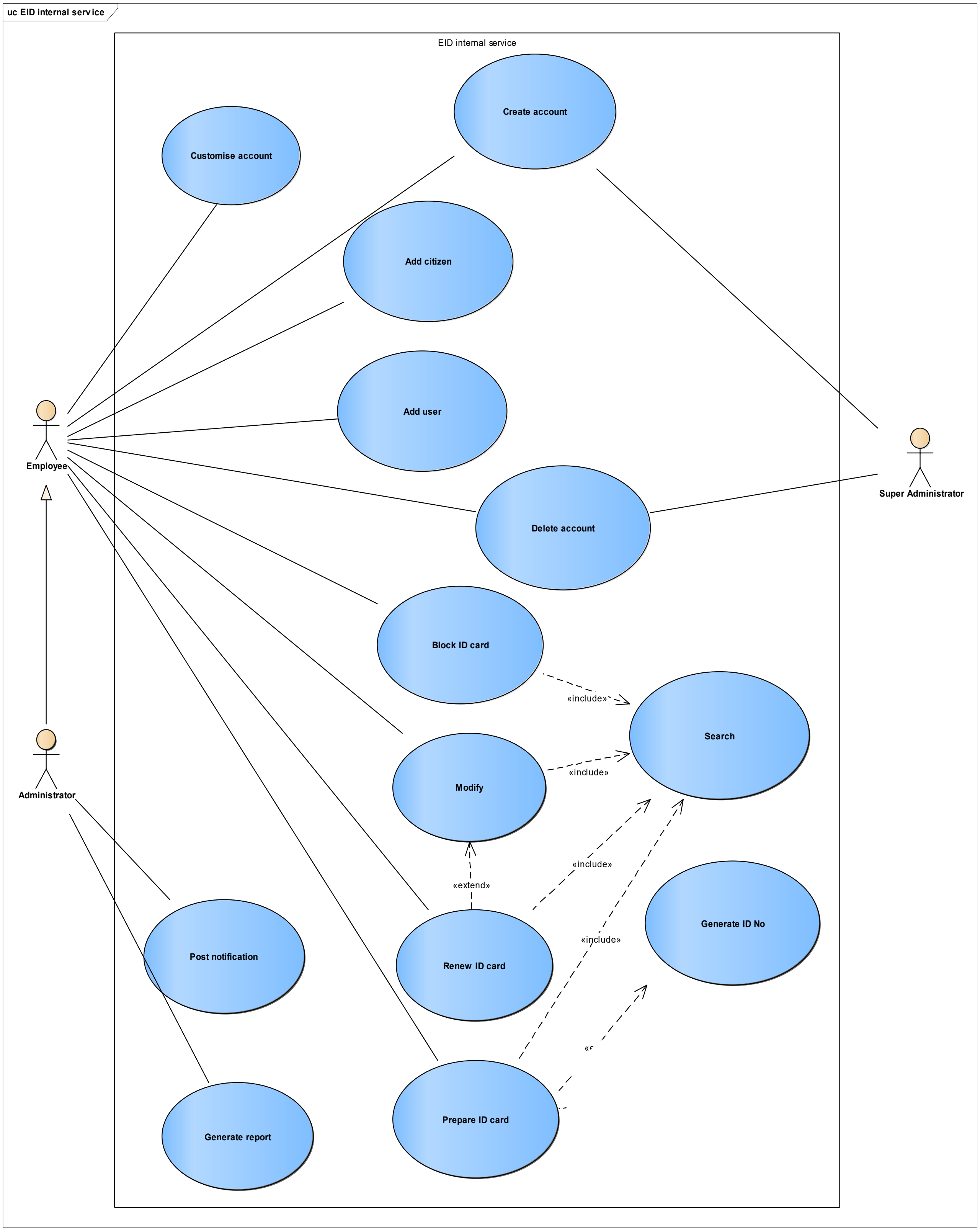

Figure 3. 3 Use case diagram for internal service

3.2.4 Use-case description

Table 3. 1 Use case description for Login

Use-case name -Login	
Use-case ID – UC 1	
Actor	• Super admin, Admin, Employee, Citizen, Organization, Police officer
Use-case description	• User of the system can login and get services from the system.
Used use-case	• Register • Create account
Pre-conditions	• Actor must have an account • Actor must have valid user-name and password
Post-condition	After successfully login an actor will be provided with it profile page and authorized access to the system
Flow of event	1. Actor opens the system (website) 2. Actor press login button 3. Actor input user name and password 4. Actor submit login 5. System authenticates user name and password 6. System loads and shows the profile page of Actor
Scenario	2a Actor has forgotten his/her password 1. Actor clicks the forget password button. 2. System sends back a temporary password to their valid contact, number 3. Actor accesses the temporary message and customizes new password 6a Incorrect user or password

	1. System displays error message 2. Actor go to step 3 and retypes user name password and login again.

Table 3. 2 Use case description for Register

Use-case name - Register	
Use-case ID – UC 2	
Actor	Employee, Admin, organization, citizen, police officer
Use case description	• Before anything happens or before any action performed, user should be registered on the system by filling necessary information. User may be asked to provide both the information and document.
Used use case	No used use-case
Pre-conditions	• Actor must have convincing document evidence for registration.
Post-condition	• Actor will successfully sent a request for registration and wait for approval and confirmation. After approval and confirmation actor should go to the governmental office for activation.
Flow of event	1. Actor opens the website. 2. Actor press registration button 3. Actor input personal information 4. Actor upload document evidence 5. System verify the personal information and the document. 6. System display message 7. System sent request for registration 8. Third party review request sent, perform document and information verification in order to accept or approve the request. Then respond to request

	9. System respond to actor
Scenario	2a Registration as Admin and Employee 1. Actor provide personal information 2. Actor upload CV and other document 2b Registration as citizen 1. Actor provide to personal information, 2. Upload birth certificate or other document evidence 2c Registration as organization 1. Actor provide branch name, address, password and name of the organization. 2. Upload license or other document evidence 6a In correct information filled 1. System display " incorrect information filled" 2. Actor go to step 3 and refill information 6b In correct document format uploaded 1. System display " document format not supported" 2. Actor go to step 4 and upload other format of document. 6c correct information filled and correct document uploaded 1. System display " Request successful sent" 8a Employee request for registration 1. Super Admin review the request 2. Super Admin perform document and information verification. 3. Super Admin respond to request. 8b Citizen, organization and police officer request for registration 1. Employee review the request

	2. Employee perform document and information verification. 3. Employee respond to request 9a Request accepted 1. System display "Request for registration has been accepted come to the office for further verification and approval". 9b Request rejected 1. System display "Request is failed you do not full fill the requirement"

Table 3. 3 Use case description for generate barcode

Use-case name - Generate barcode	
Use-case ID – UC 3	
Actor	Admin, Employee
Use-case description	• Barcode is used to hold some data attributes. The ID card of citizens has barcode. First of all the barcode has to be generated based on the required attribute or parameter. Then the generated bar code can be putted on citizen ID card.
Used use-case	• Login • Search
Pre-conditions	• Citizens must be registered before barcode or ID card is assigned to them
Post-condition	• unique barcode Generated and assigned to ID card or citizen

Flow of event	1. Actor open website 2. Actor press login button 3. Actor fill user-name and password 4. Actor submit login 5. System provide Actor's profile page 6. Actor Search target citizen by filling citizen information 7. Actor submit search 8. System fetch and display possible citizens as a result of searching 9. Actor choose target citizen and open its profile 10. Actor click generate barcode button 11. System display "choose" parameter (attribute) 12. Actor choose parameters (attributes) 13. The system generates unique barcode using the selected parameters 14. Actor integrate bar code with ID cards
Scenario	6b Search using Registration number 1. Actor input registration ID number 6c Search using Name 1. Actor enter citizen ID number

Table 3. 4 Use case description for Prepare ID card

Use-case name - Prepare ID card	
Use-case ID – UC 4	
Actor	Employee , Admin
Use-case description	• Citizen who are above 18 years of age may have ID card. ID card should be prepared for those citizens. System fetch citizen information from database and

	organize the data the put necessary information that should be displayed on citizen ID cards. This use case allows the employee or the admin to prepare ID card for the citizen if he or she is already registered on the database or have a justification to have ID in other way.
Used use-case	• Login • Search • Generate barcode • Generate ID number
Pre-conditions	• The citizen whom employee going to prepare ID card must be registered in the database
Post-condition	ID card will be prepared with unique ID number and unique barcode
Flow of event	1. Actor open website 2. Actor press login button 3. Actor fill user-name and password 4. Actor submit login 5. System provide Actor's profile page 6. Actor Search target citizen by filling citizen information 7. Actor submit search 8. System fetch and display possible citizens as a result of searching 9. Actor choose target citizen and open its profile 10. Actor press prepare ID card 11. System fetch citizen information from database 12. System organize the information that is going to be putted on the card 13. System prepare ID card 14. Actor press display ID card button 15. System preview ID card 16. Actor press print ID card button 17. System print ID card

Alternative actions	12 a Target citizen already have card ID number 1. Actor press" add ID card number" 2. System integrate ID card number to ID cards 3. Actor press " add barcode" 4. System integrate barcode to ID card 12b Target citizen do not have card ID number 1. Actor press "Generate ID card number" 2. System generate unique ID card number 3. Actor press " add ID card number" 4. System integrate ID card number to ID card 5. Actor press "Generate barcode" 6. System generate unique barcode 7. Actor press " add barcode" 8. System integrate barcode with ID cards

Table 3. 5 Use case description for Generate ID card number

Use-case name - Generate ID card number	
Use-case ID – UC 5	
Actor	Admin , Employee
Use-case description	• ID card number use to identify citizen who have ID card. The ID card of citizens has ID number. First of all the ID number has to be generated based on the. Then the generated ID number can be putted on citizen ID card.
Used use-case	• Login • Search

Pre-conditions	✓ Actors must have registered on the system. ✓ Actors must have prepared ID card
Post-condition	Unique ID number is generated and assigned to ID card or citizen
Flow of event	1. Actor open website 2. Actor press login button 3. Actor fill user-name and password 4. Actor submit login 5. System provide Actor's profile page 6. Actor Search target citizen by filling citizen information 7. Actor submit search 8. System fetch and display possible citizens as a result of searching 9. Actor choose target citizen and open its profile 10. Actor press button generate ID card number 11. System generate unique ID card number 12. Actor integrate ID number with ID card
Scenario	6a Search using registration number 1. Actor input registration ID number 6b Search using Name 1. Actor input citizen name

Table 3. 6 Use case description for Modify

Use-case name - Modify	
Use-case ID – UC 6	
Actor	Employee , Admin

Used use-case	• Login • search
Pre-conditions	• Actor should upload document evidence for reason to modification
Post-condition	Profile will be set or changed by the Employee and Admin
Flow of event	1. Actor open website 2. Actor press login button 3. Actor fill user-name and password 4. Actor submit login 5. System provide Actor's profile page 6. Actor Search target citizen by filling citizen information 7. Actor submit search 8. System fetch and display possible citizens as a result of searching 9. Actor choose target citizen and open its profile 10. Actor press modify button. 11. Actor enter the data to be modified 12. Actor submit modification 13. System verify the entered data 14. System perform modification
Scenario	6a Search using barcode • Actor put citizen ID card to barcode reader • The reader read the barcode • System fetch the code (ID number or other parameter) from the reader 6b Search using ID number 2. Actor enter citizen ID number 6c Search using Name

	2. Actor enter citizen ID number
Exception	✓ Incorrect citizen identified or un registered citizen ✓ Unavailable barcode number

Table 3. 7 Use case description for Customize account

Use-case name - Customize account	
Use-case ID – UC 7	
Actor	Admin , Employee ,legal body ,citizen, organization
Used use-case	Login
Pre-conditions	Actor must have user name and password
Post-condition	Account will be changed or modified by Actor
Flow of event	1. Actor open website 2. Actor press login button 3. Actor fill user-name and password 4. Actor submit login 5. System provide Actor's profile page 6. Actor press "Setting" button. 7. Actor change user name 8. Actor change password 9. Actor submit changes 10. System perform the changes 11. System re-direct to actor's customize-page displayed 12. System registers change of profile.

Exception	✓ un registered citizen

Table 3. 8 Use case description for Request for ID cards

Use-case name - Request for ID cards	
Use-case ID – UC 8	
Actor	Citizen
Use-case description	• Citizen can request for ID card online. Citizen who are above 18 years of age can have ID cards. To get this ID card they can request for it online then they can take it by going to IDCO (Identification card office or center).
Used use-case	Login
Pre-conditions	• Actor must have user name and password • Actor must be registered
Post-condition	Request will be sent by Actor.

Flow of event	1. Actor open website 2. Actor press login button 3. Actor fill user-name and password 4. Actor submit login 5. System provide Actor's profile page 6. Actor clicks "request ID card" button. 7. System ask reason 8. Actor choose reason 9. Actor upload document evidences 10. Actor submit request 11. System display confirmation message 12. System display meeting time for approval and ID card taking
Scenario	8a Already having ID cards 1. Actor choose " existing ID card" 8b Have no any ID cards before 1. Actor choose "new ID card"
Exception	

Table 3. 9 Use case description for Display

Use-case name - Display	
Use-case ID – UC 9	
Actor	Organization, policeman, employees, admin, and police officer an ordinary citizen
Use case description	This use case allows actors (employee, registered organization, police, and others if any) to see information of the citizen based on their need.

Used use-case	• Login • Search
Pre-conditions	• Any actor whether organization or person should have permission to get citizens information, and can also get the only information that is necessary and allowable to it.
Post-condition	Employee or admin will have clear understanding or information about the citizen and can take their decision according to the law.
Flow of event	1. Actor open website 2. Actor press login button 3. Actor fill user-name and password 4. Actor submit login 5. System provide Actor's profile page 6. Actor Search target citizen by filling citizen information 7. Actor submit search 8. System fetch and display possible citizens as a result of searching 9. Actor choose target citizen and open its profile 10. Actor press view citizen information 11. System display citizen information
Scenario	6a Search using barcode 1. Actor put citizen ID card to barcode reader 2. The reader read the barcode 3. System fetch the code (ID number or other parameter) from the reader 6b Search using ID number 1. Actor enter citizen ID number 6c Search using Name 1. Actor enter citizen ID number

Table 3. 10 Use case description for verify ID card

Use-case name - Verify ID card	
Use-case ID – UC 10	
Actor	Organization, policeman, employees, admin, and police officer an ordinary citizen
Use-case description	• This use-case allow different user to check the validity of citizen ID card. User can check whether an ID card is legal, valid or original.
Used use-case	• Login • Search
Pre-conditions	• Actor must have permission and have to be authenticated user name and password.
Post-condition	Actor will verify the ID card of the citizen
Flow of event	1. Actor open website 2. Actor press login button 3. Actor fill user-name and password 4. Actor submit login 5. System provide Actor's profile page 6. Actor Search target citizen by filling citizen information 7. Actor submit search 8. System fetch and display possible citizens as a result of searching 9. Actor choose target citizen and open its profile 10. Actor press verify citizen ID card 11. System verify ID card 12. System Display message

Scenario	6a Search using barcode 1. Actor put citizen ID card to barcode reader 2. The reader read the barcode 3. System fetch the code (ID number or other parameter) from the reader 6b Search using ID number 1. Actor enter citizen ID number 6c Search using Name 1. Actor enter citizen ID number 8a Un known ID card(Target citizen not found) 1. System display “invalid and unknown ID card”. 12b Known but out of date ID card 1. System display ”ID card is out of data must be renewed” 12c valid and known ID card 1. System display “Valid ID card”

Table 3. 11 Use case description for Block ID card

Use-case name - Block ID card	
Use-case ID – UC 11	
Actor	Employee and admin
Use-case description	• Make given ID card out of service. If ID cards are banned they are invalid and can cannot be used as legal card.
Use case	This use case allows the employee or admin of the system to block user

description	accounts if the user does not use the system appropriately or if his or her id is not renewed.
Used use-case	• Login • Search
Pre-conditions	There should be enough reason for blocking ID card. And it should be supported bay document evidences.
Post-condition	• The citizen ID card become useless • If it not fair the actor has to take measure to undo it.
Flow of event	1. Actor open website 2. Actor press login button 3. Actor fill user-name and password 4. Actor submit login 5. System provide Actor's profile page 6. Actor Search target citizen by filling citizen information 7. Actor submit search 8. System fetch and display possible citizens as a result of searching 9. Actor choose target citizen and open its profile 10. actor press Block ID card button 11. System ask reason and information and document evidence 12. Actor fill reason , information and upload document 13. Actor submit it 14. System will block that target ID Card
Scenario	6a Search using barcode 1. Actor put citizen ID card to barcode reader 2. The reader read the barcode 3. System fetch the code (ID number or other parameter) from the reader

	6b Search using ID number 1. Actor enter citizen ID number 6c Search using Name 1. Actor enter citizen ID number

Table 3. 12 Use case description for post notification

Use-case name - Post notification	
Use-case ID – UC 12	
Actor	Admin, Super admin
Used use-case	Login
Pre-conditions	✓ Actor must have an account and must add one or more employees before posting notifications to employees. ✓ Employee must register one or more citizens to system.
Post-condition	After posting notifications to employees, Admin can see responses from employees.
Flow of event	1. Actor open website 2. Actor press login button 3. Actor fill user-name and password 4. Actor submit login 5. System provide Actor's profile page 6. Actor press the post notification button. 7. Actor writes notification message in text area. Then 8. Actor selects employees/citizens of which the message is concerned. 9. Actor send the message 10. System broadcasts Actor's message to client (employees and citizens).

Scenario	9a Actor sends incorrect notification to employees. 1. Actor edit the message 2. Actor resend it. 3. System broadcast the message to appropriate client 9b Actor sends notification to wrong employee. 1. Actor deletes the message and 2. Actor sends description message to employee who accepts undue message.

Table 3. 13 Use case description for Renewing ID card

Use-case name - Renew ID Card	
Use-case ID – UC 13	
Actor	Employee, Admin
Use case description	This use case allows the employee or admin to renew citizen's id.
Used use-case	• Login • Search
Pre-conditions	• The system should send notification as the expired date of the ID card is reached • The citizen should be presented physically in the employee office
Post-condition	• The ID card of the citizen is renewed • The timer starts to count from this time in order to set the expired date of the ID card for the next time.
Flow of event	1. Actor open website 2. Actor press login button

	3. Actor fill user-name and password 4. Actor submit login 5. System provide Actor's profile page 6. Actor Search target citizen by filling citizen information 7. Actor submit search 8. System fetch and display possible citizens as a result of searching 9. Actor choose target citizen and open its profile 10. actor update necessary information of target citizen 11. Actor upload document evidences 12. Actor submit modification 13. System restart timer 14. System ask the actor whether to reprint ID card or not 15. Actor response 16. System perform some action according to actor's response.
Scenario	15a Information in ID card changed 1. Actor choose reprint ID card 2. System print ID card of target citizen 15b Information in ID card not changed 1. Actor choose do not reprint ID card

Table 3. 14 Use case diagram for search

Use-case name - Search	
Use-case ID – UC 14	
Actor	Admin, organization, police officer, Employee, citizen and super admin
Use-case description	• Search user by entering different information.eg search using name and search using ID number.

Used use-case	Login
Pre-conditions	Actor must login successfully to the system.
Post-condition	Actor can search any data in the system based on their scope of access and see the result.
Flow of event	1. Actor open website 2. Actor press login button 3. Actor fill user-name and password 4. Actor submit login 5. System provide Actor's profile page 6. Actor Search target citizen by filling citizen information 7. Actor submit search 8. System fetch and display possible citizens as a result of searching 9. Actor choose target citizen and open its profile
Scenario	6a Search using barcode • Actor put citizen ID card to barcode reader • The reader read the barcode • System fetch the code (ID number or other parameter) from the reader 6b Search using ID number 3. Actor enter citizen ID number 6c Search using Name 3. Actor enter citizen ID number
Exception	Actor did not find the data or target citizen after searching.

Table 3. 15 Use case description for Set citizen as wanted

Use-case name - Set citizen as wanted	
Use-case ID – UC 15	
Actor	Employee, Admin
Use-case description	• This use- case make citizen wanted or add an information to the database that citizens are wanted. And any one seeing those suspect citizens can report to the police. If they are wanted the system display that they are wanted.
Use-case description	This use case allows the employee or admin to make citizens wanted accordingly.
Used use-case	Login
Pre-conditions	The employee or admin should have enough reason to make citizen wanted or there should be request from the police officer.
Post-condition	Citizen will be wanted, and this will be added to his data.
Flow of event	1. Actor open website 2. Actor press login button 3. Actor fill user-name and password 4. Actor submit login 5. System provide Actor's profile page 6. Actor Search target citizen by filling citizen information 7. Actor submit search 8. System fetch and display possible citizens as a result of searching 9. Actor choose target citizen and open its profile 10. Actor press make citizen wanted button 11. System ask reason 12. Actor fill reason 13. System ask actor to upload document evidence

	14. Actor upload document 15. System make citizen wanted and by setting flag for wanted.
Scenario	6a Search using barcode 1. Actor put citizen ID card to barcode reader 2. The reader read the barcode 3. System fetch the code (ID number or other parameter) from the reader 6b Search using ID number 1. Actor enter citizen ID number 6c Search using Name 1. Actor enter citizen ID number

Table 3. 16 Use case description for setting service ban

Use-case name - Ban service	
Use-case ID – UC 16	
Actor	Employee and Admin
Use-case description	• Used to ban citizen from different services. So that if citizen are banned they cannot have the banned service using their ID card.
Use case description	This use case allows the employee or admin to ban citizens from some service accordingly.
Used use-case	login
Pre-conditions	The employee or admin should have enough reason to ban citizen from services

Post-condition	The citizen will be banned from some services based on their case
Flow of event	1. Actor open website 2. Actor press login button 3. Actor fill user-name and password 4. Actor submit login 5. System provide Actor's profile page 6. Actor Search target citizen by filling citizen information 7. Actor submit search 8. System fetch and display possible citizens as a result of searching 9. Actor choose target citizen and open its profile 10. Actor press ban service button 11. System ask type of service the target citizen is going to be ban from 12. System ask reason 13. Actor fill reason 14. System ask actor to upload document evidence 15. Actor upload document 16. System ban citizen from services and by setting flag for wanted.
Scenario	6a Search using barcode 1. Actor put citizen ID card to barcode reader 2. The reader read the barcode 3. System fetch the code (ID number or other parameter) from the reader 6b Search using ID number 1. Actor enter citizen ID number 6c Search using Name 1. Actor enter citizen ID number

Table 3. 17 Use case description for View report

<table>
<tr><td colspan="2">Use-case name - View report</td></tr>
<tr><td colspan="2">Use-case ID – UC 17</td></tr>
<tr><td>Actor</td><td>Admin , super admin</td></tr>
<tr><td>Use-case description</td><td>This use-case enable the admin to see transaction and statistical report.</td></tr>
<tr><td>Used use-case</td><td>Login</td></tr>
<tr><td>Pre-conditions</td><td>Actor must successfully login in order to send report.</td></tr>
<tr><td>Post-condition</td><td>Once an actor successfully accesses the system, it can view statistical and transaction report.</td></tr>
<tr><td>Flow of event</td><td>Actor open website
Actor press login button
Actor fill user-name and password
Actor submit login
System provide Actor’s profile page
Actor request the report page
System provide the report page
Actor view report</td></tr>
<tr><td>Scenario</td><td>8a Transaction report
Actor ask transactions made, type, time and case by the team of employee
System display transaction detail using charts
8b Statistical Report
Actor ask total number of users and citizens</td></tr>
</table>

	2. System display total number of users and citizens using charts

3.3 Dynamic modeling

3.3.1 Activity Diagram

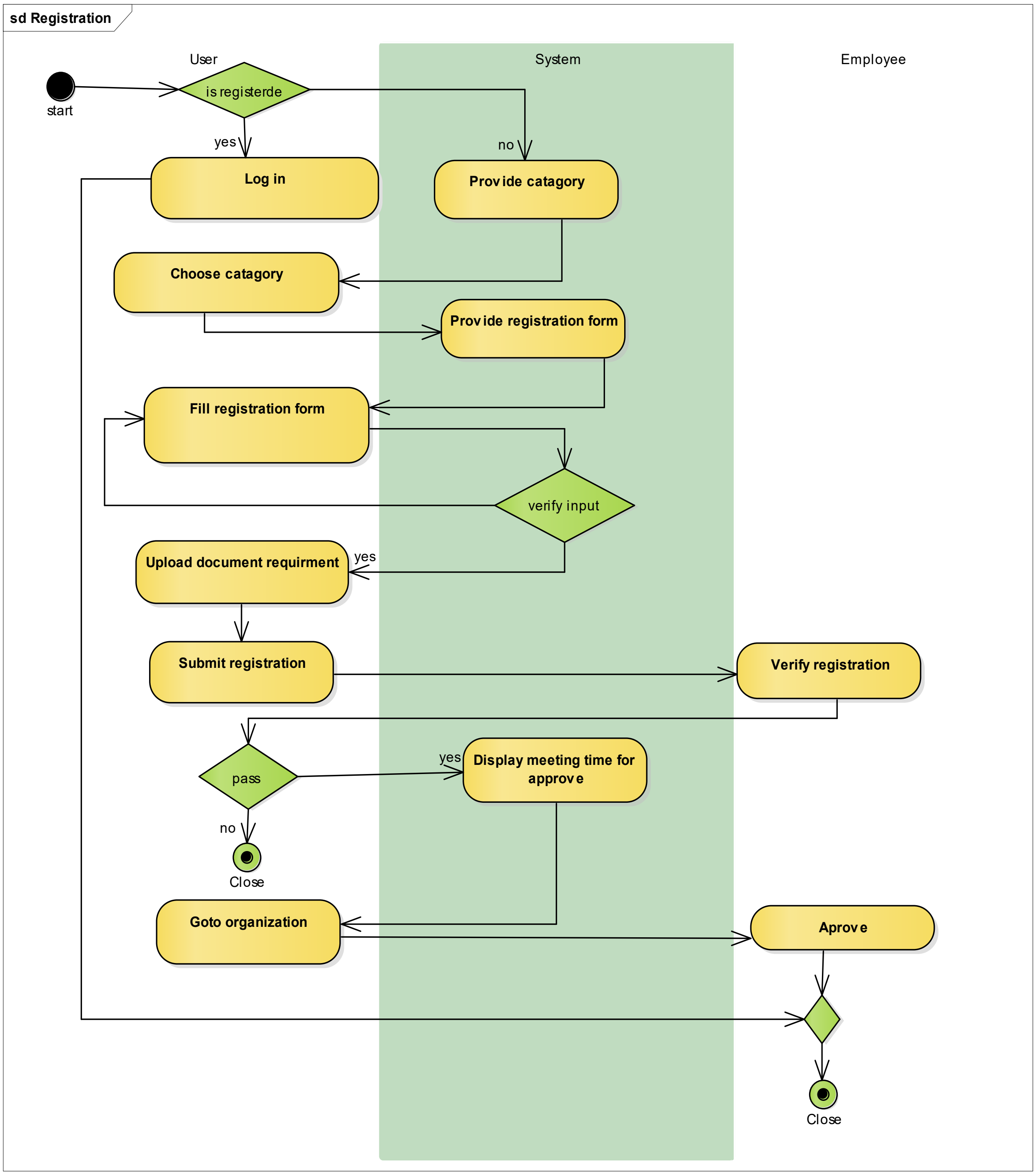

Figure 3. 4 Activity diagram for registration

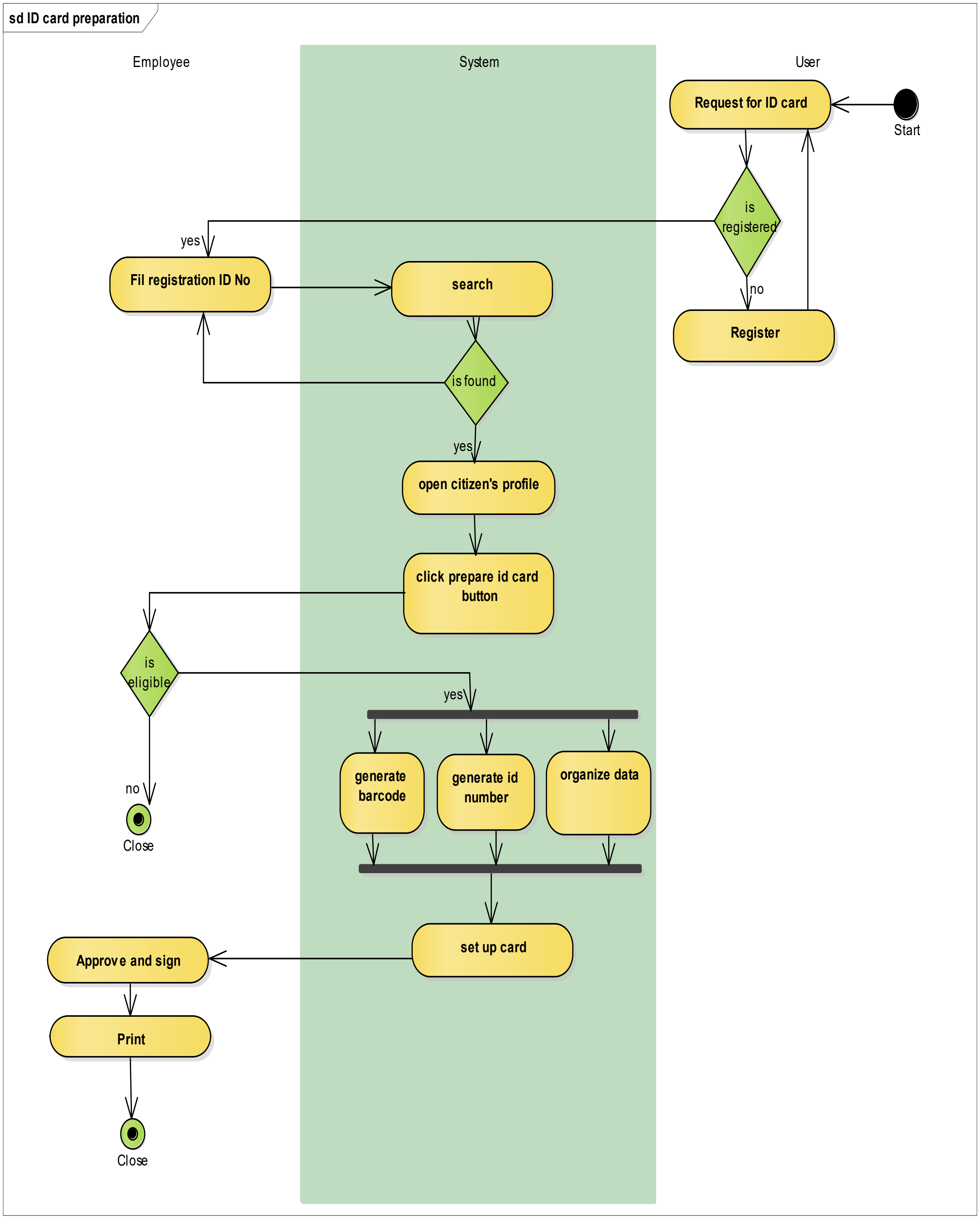

Figure 3. 5 Activity diagram for ID card preparation

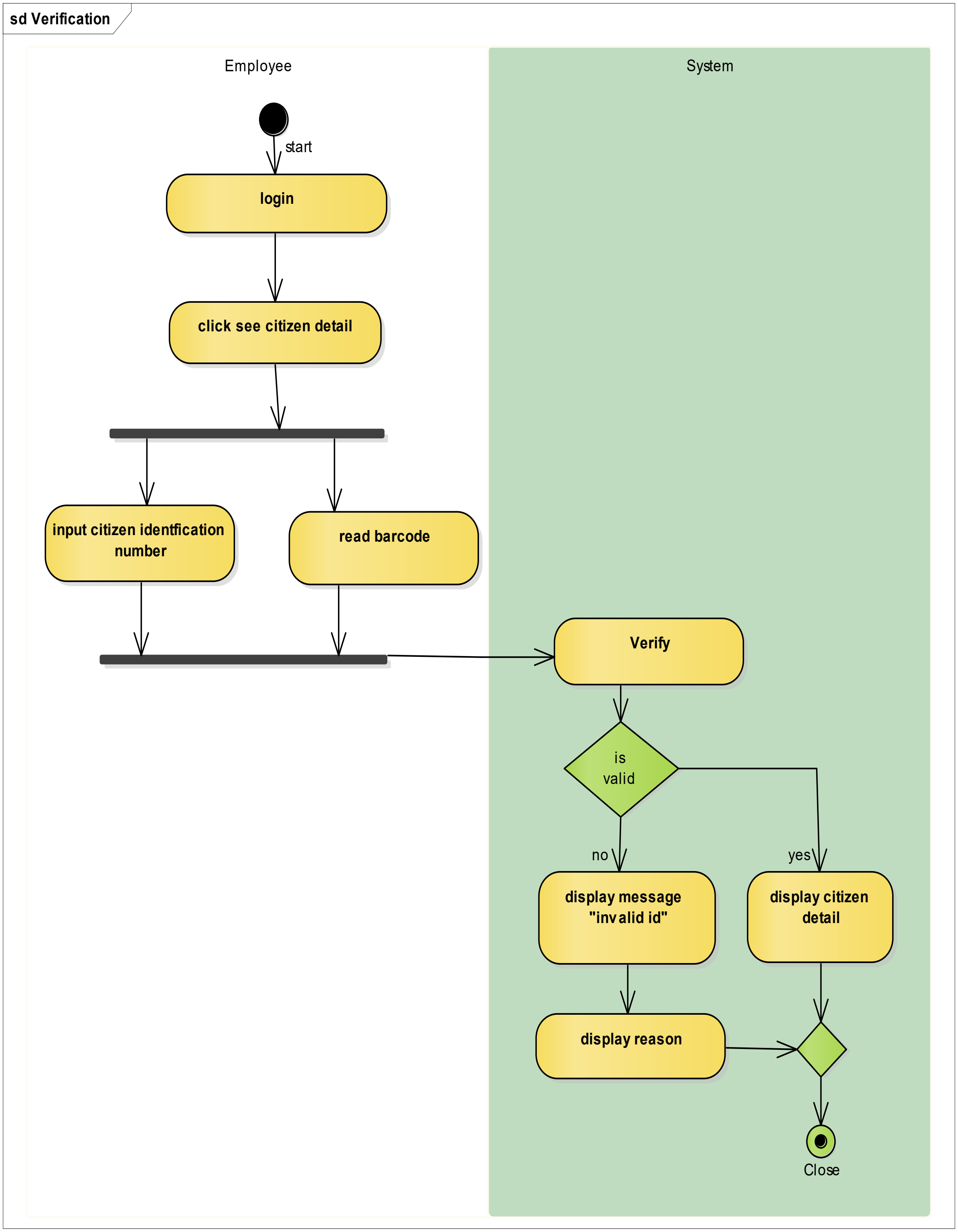

Figure 3. 6 Activity diagram for ID card verification and displaying citizen data

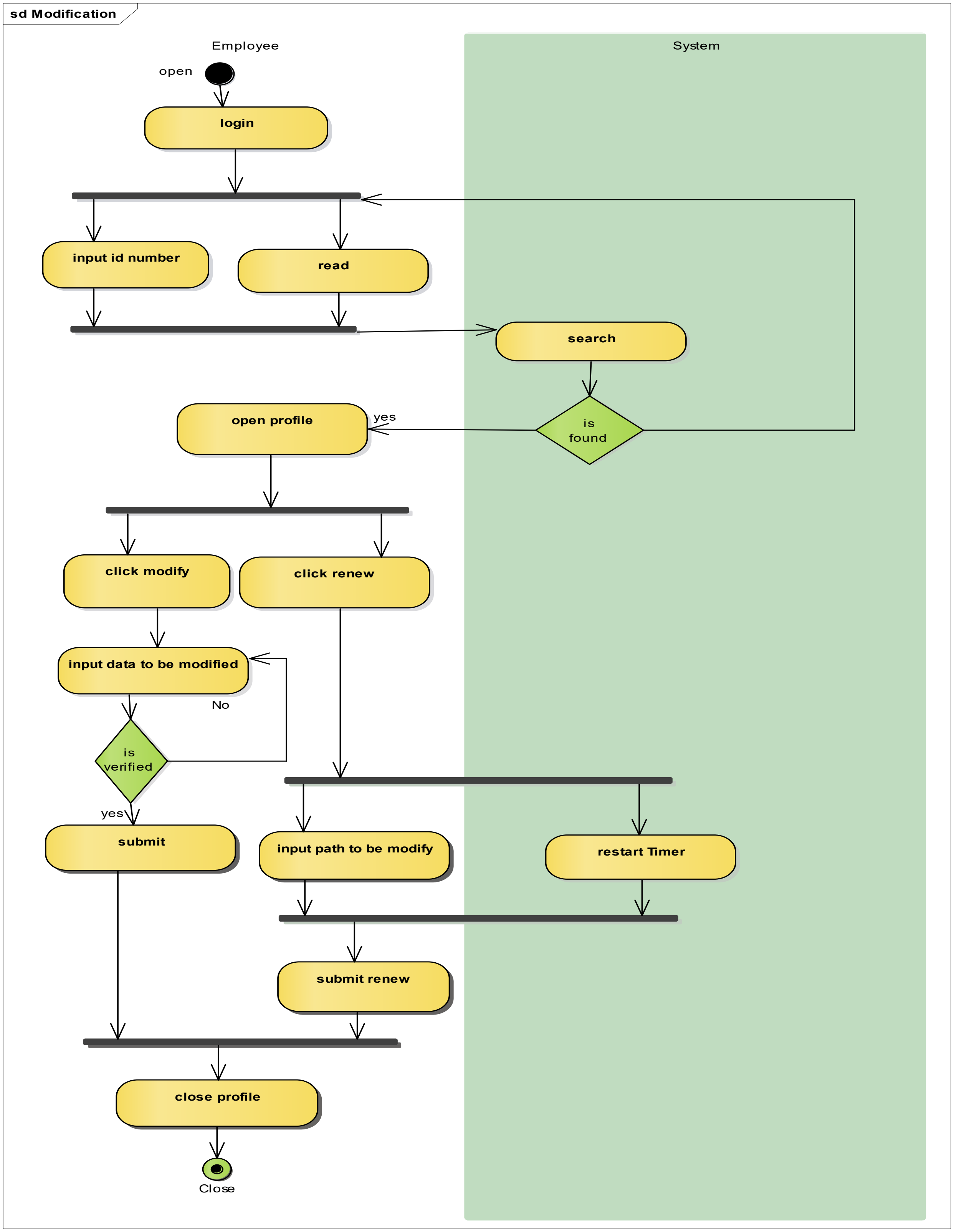

Figure 3. 7 Activity diagram for renewing ID card and modifying citizen data

3.3.2 Sequence Diagram

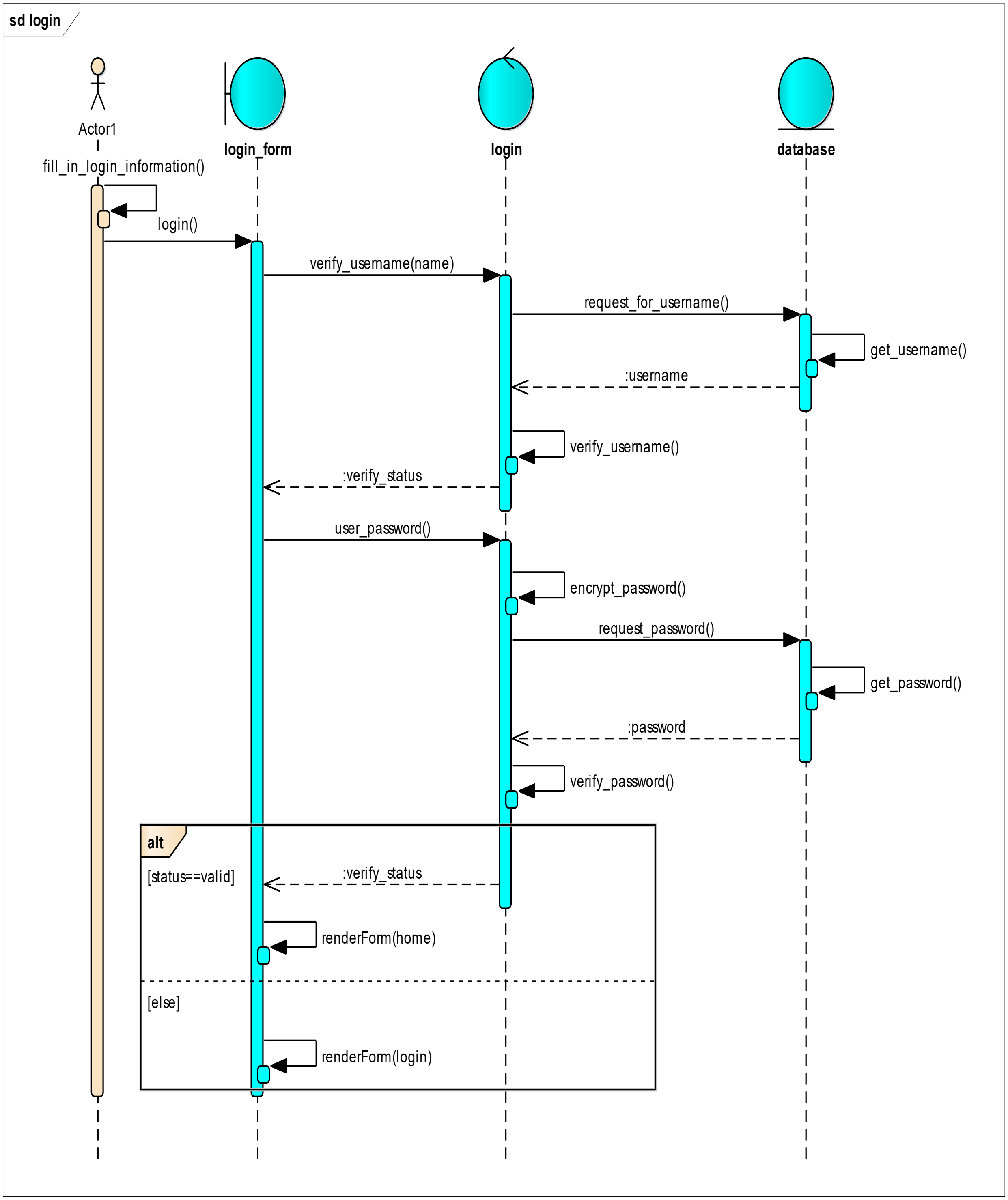

Figure 3. 8 Sequence diagram for Login

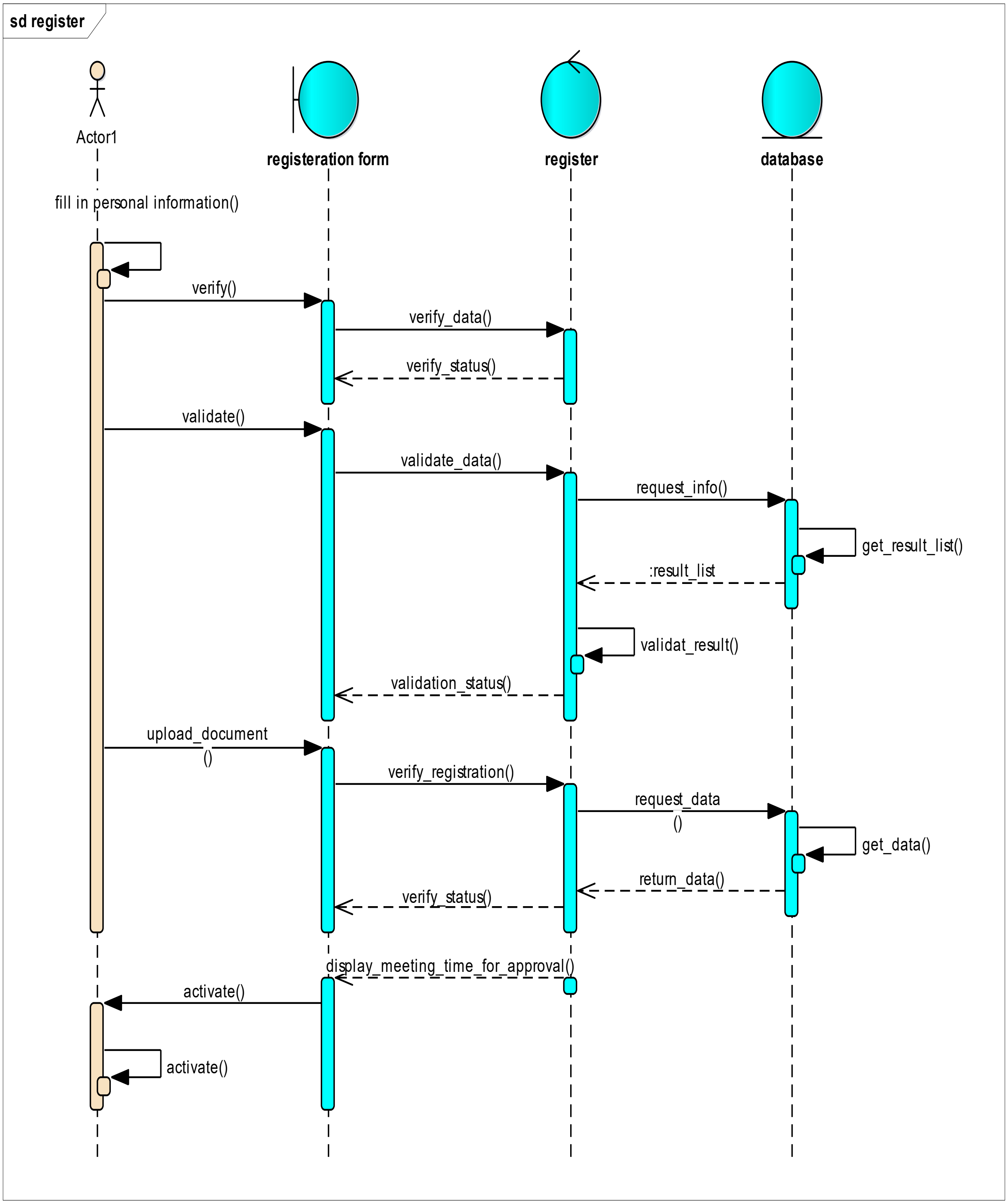

Figure 3. 9 Sequence diagram for registering

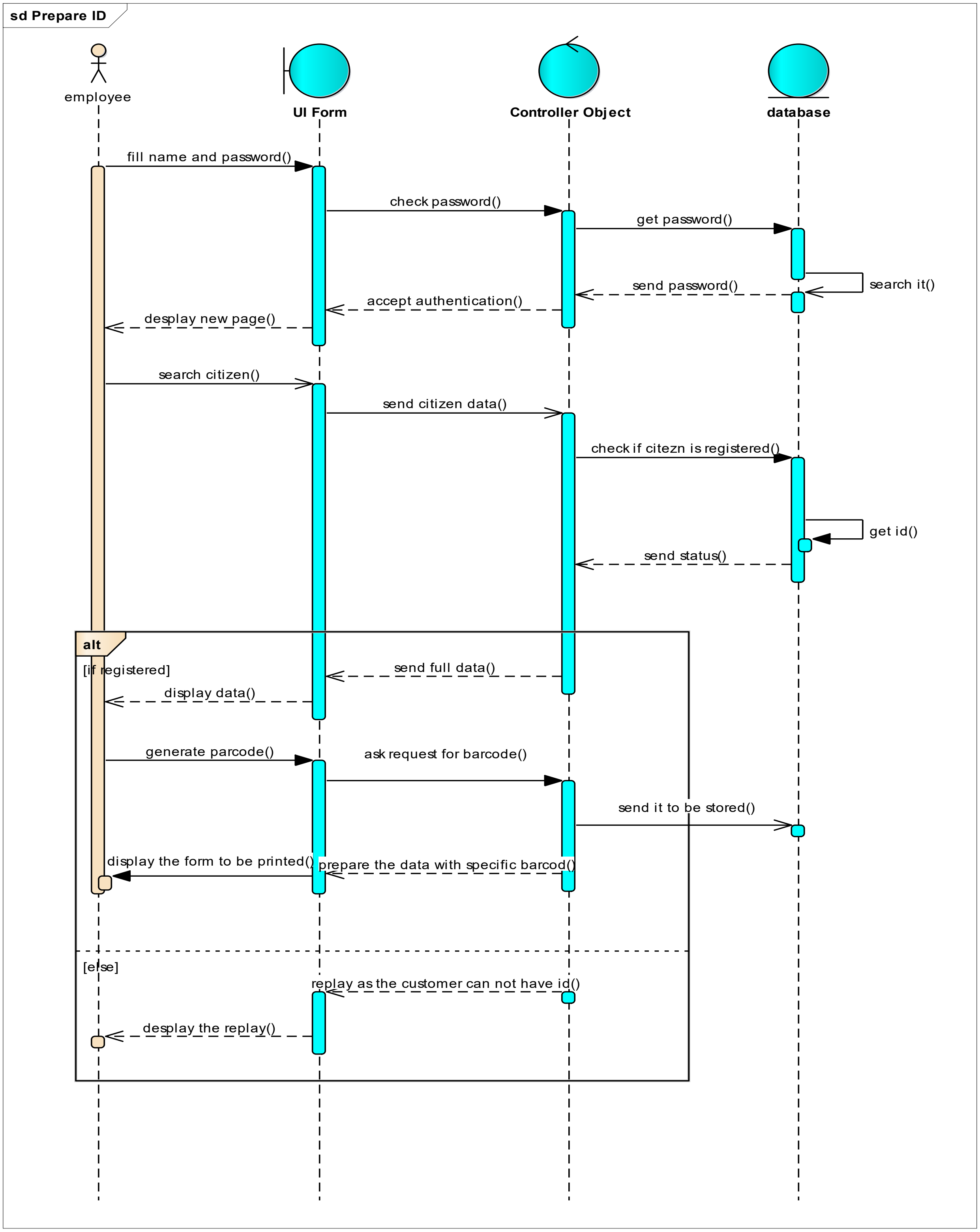

Figure 3. 10 Sequence diagram for preparing ID card

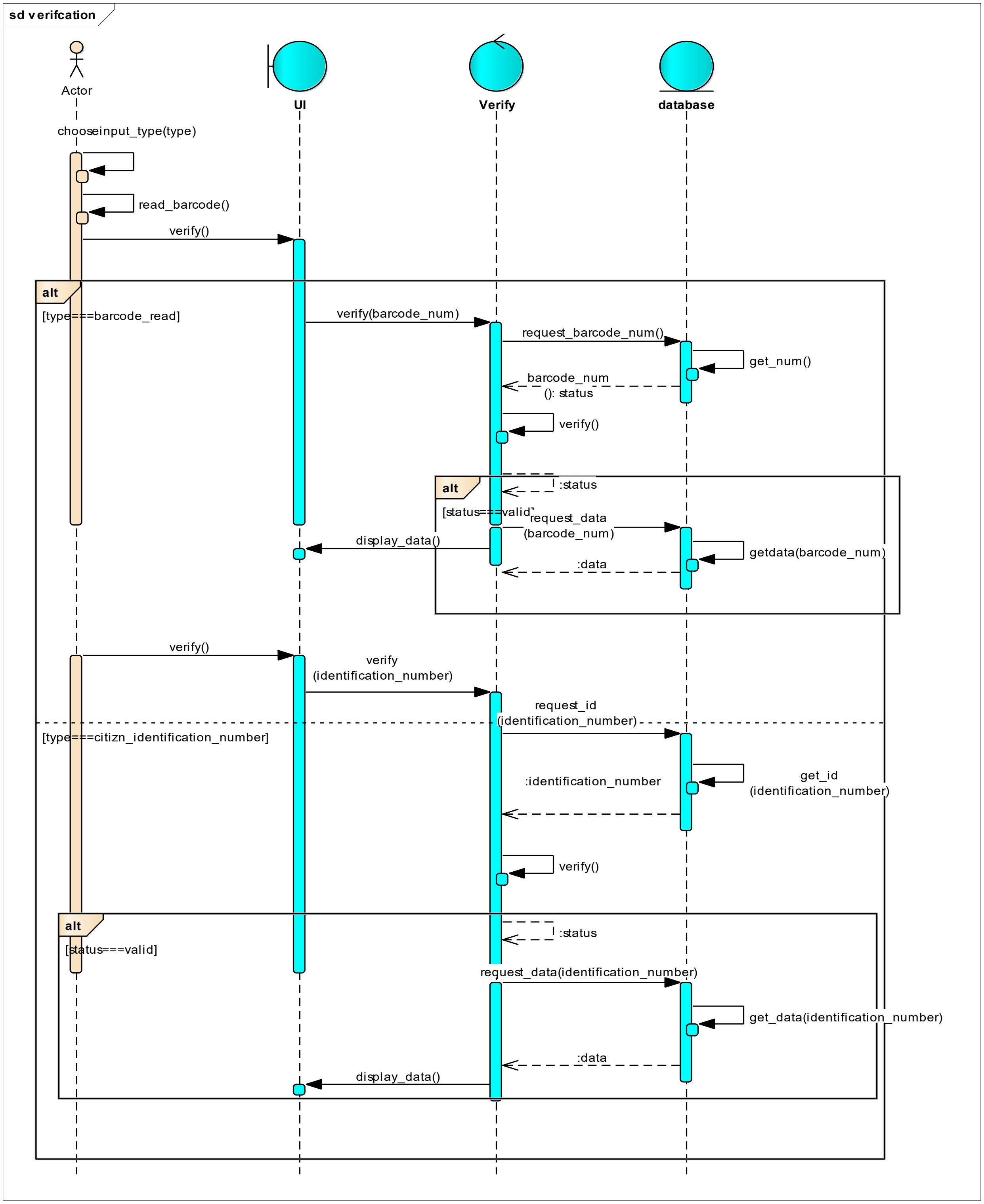

Figure 3. 11 Sequence diagram for verifying ID card and viewing citizen information

3.3.3 State Diagram

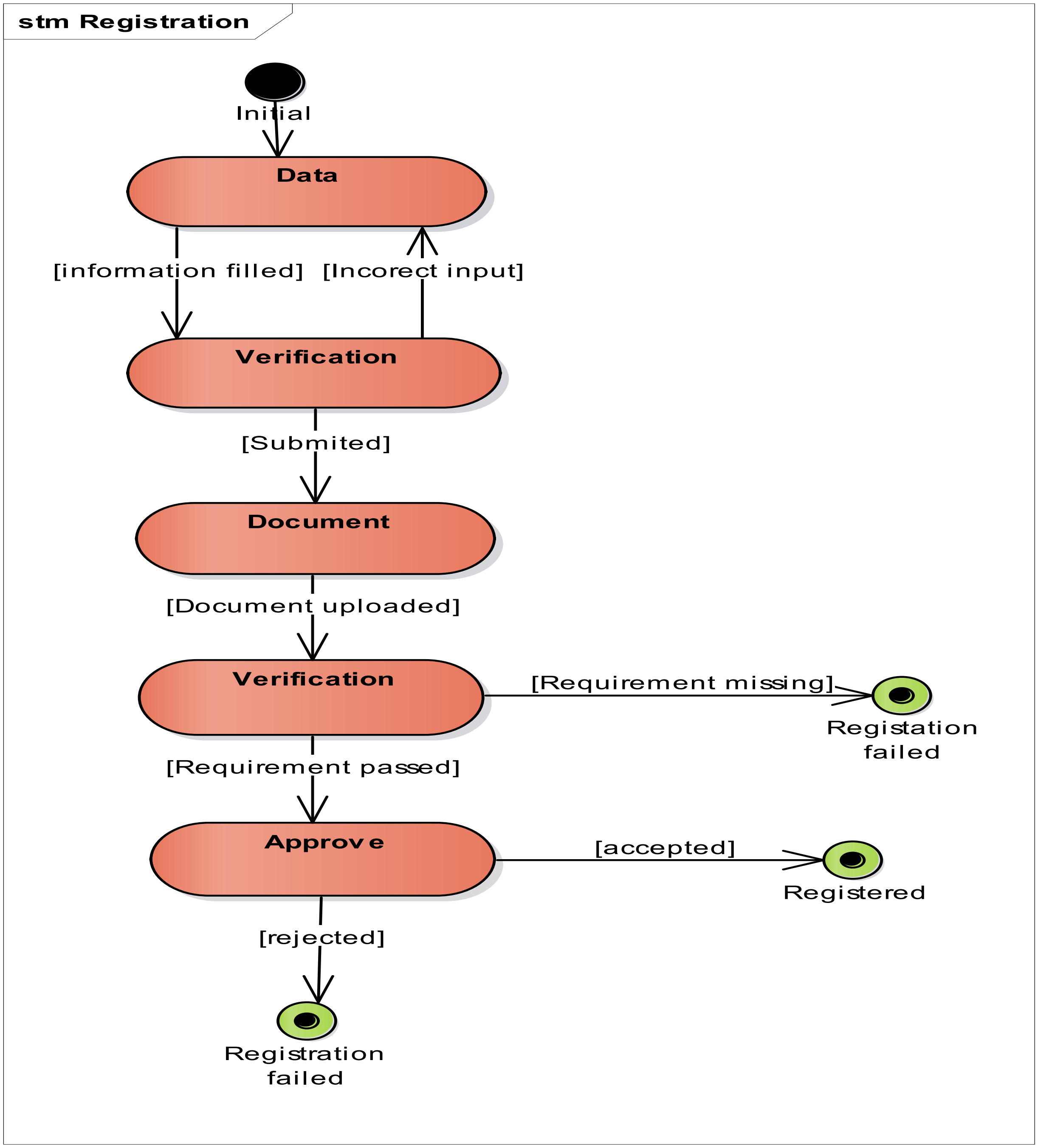

Figure 3. 12 State diagram for Registration

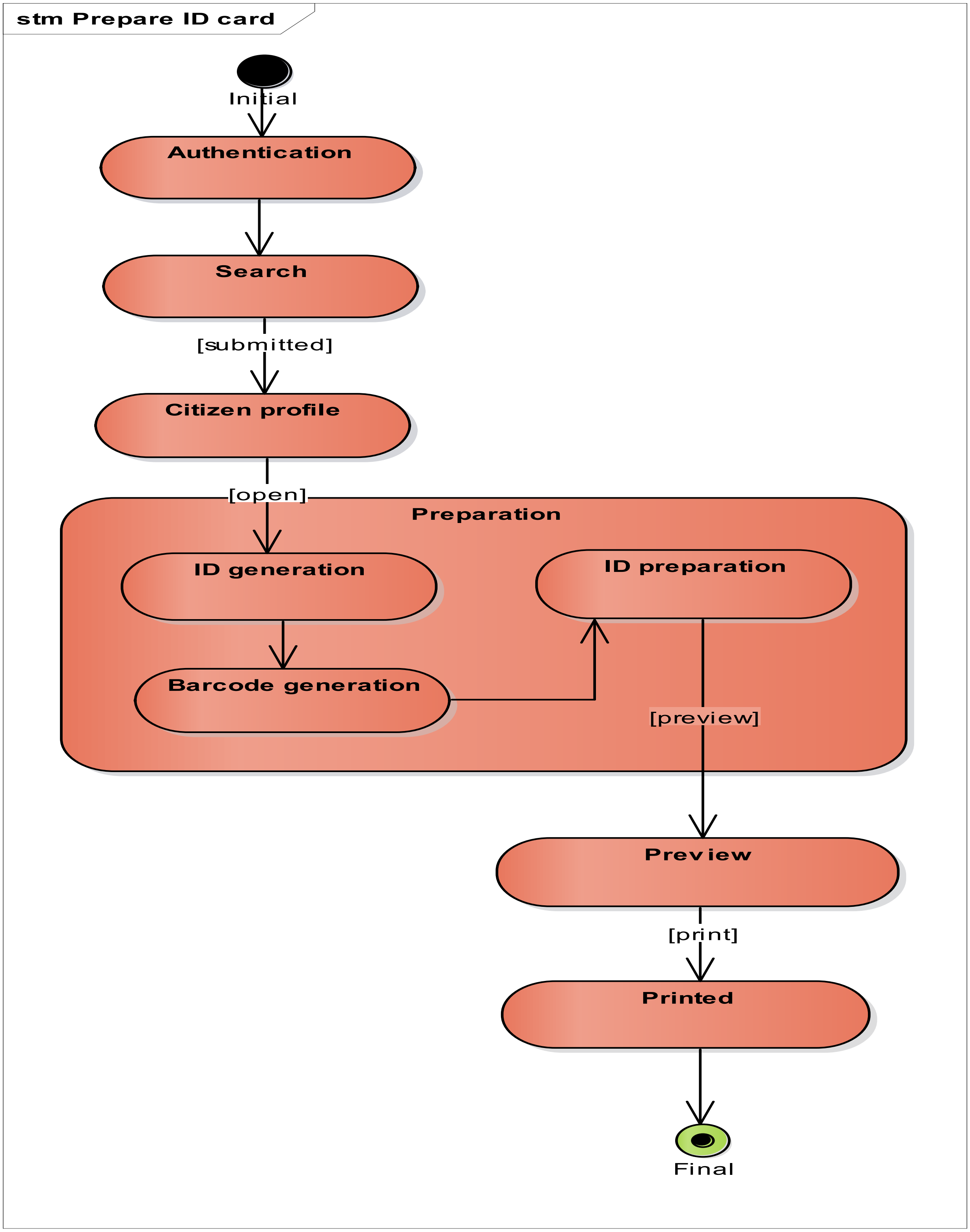

Figure 3. 13 State diagram for ID card preparation

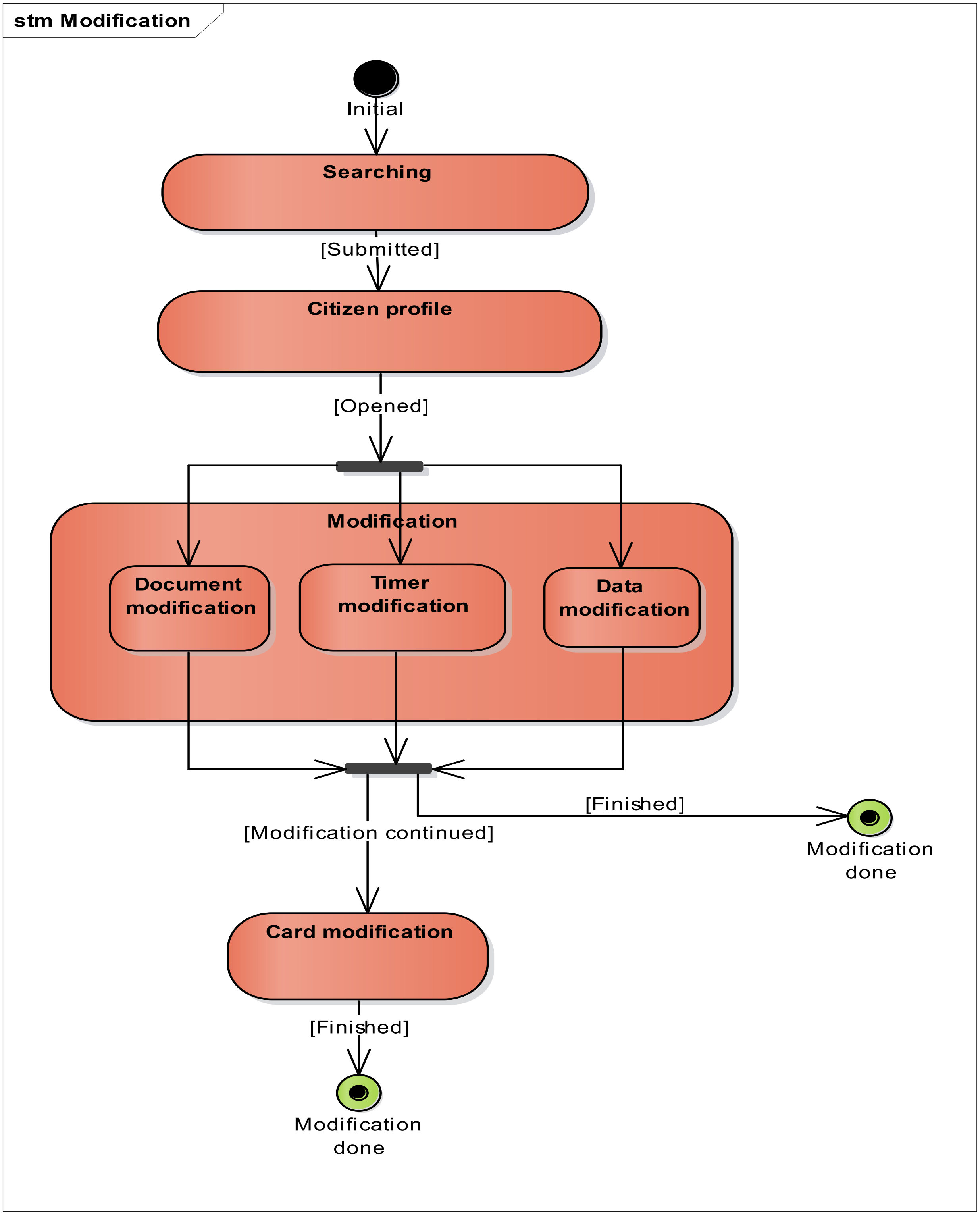

Figure 3. 14 State diagram for ID card renewal

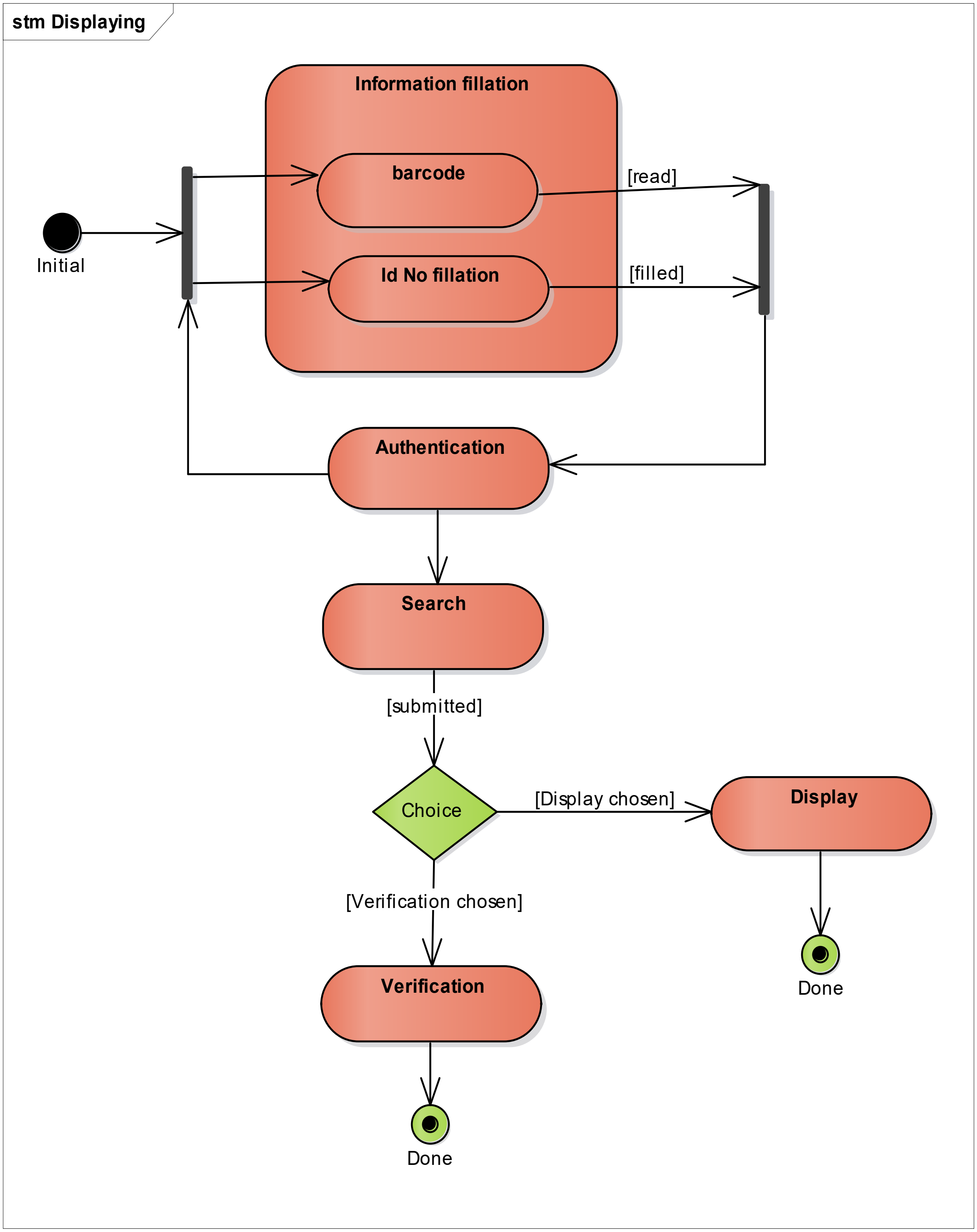

Figure 3. 15 State diagram for verifying and displaying citizen information

3.4 Class-base modeling

3.4.1 Class Identification

- User
- Citizens
- Organization
- Employee
- Admin
- Police
- Registration
- Notification
- User service
- Citizen Service
- ID Card
- Report
- Account
- Super Admin
- Transaction

3.4.2 Class Diagram

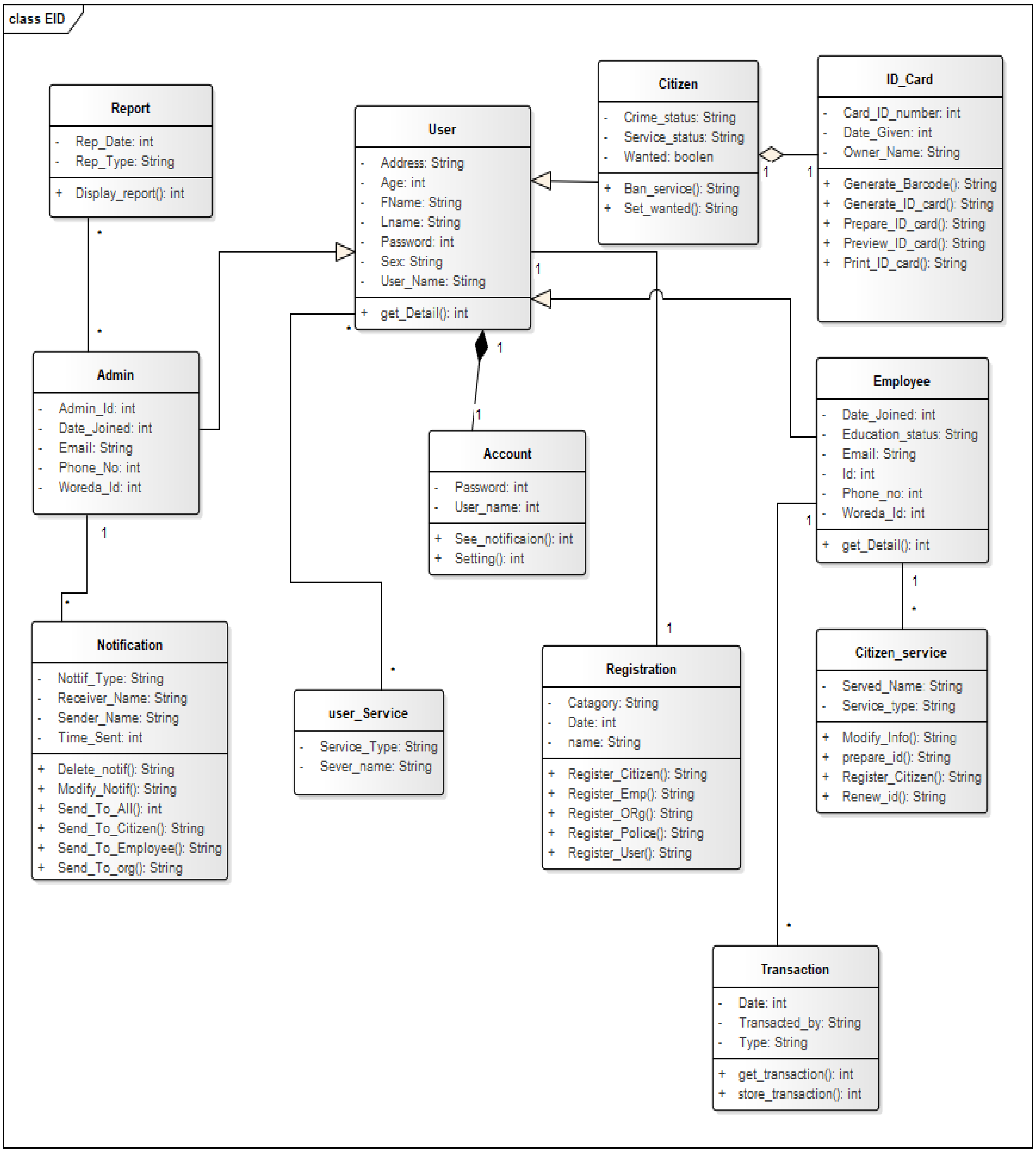

Figure 3. 16 Class diagram of ECIDS

CHAPTER FOUR: System Design

4.1 Overview

In this section we define the system design of our system. Our design includes decomposing system to relatively small components in which elements in the components are categorized by many factors such as similarities in function, have difference in function but having one objective and others factors. Our design also defines the architecture design, interface design, database design. This section provide detail information about the system and system design this help the system we are going to develop, fulfill all requirement and gives direction or roadmap to the implementation or coding phase. The detail information is supported by various diagrams such as component diagram, deployment diagram a class diagram and ER (Entity relationship) diagram.

4.2 System Design

4.2.1 System Decomposition

Our system is decomposed to modules namely User of system, ID card, Verify, View Detail, Update, Report, Utility. These modules are described in section 4.4.2.

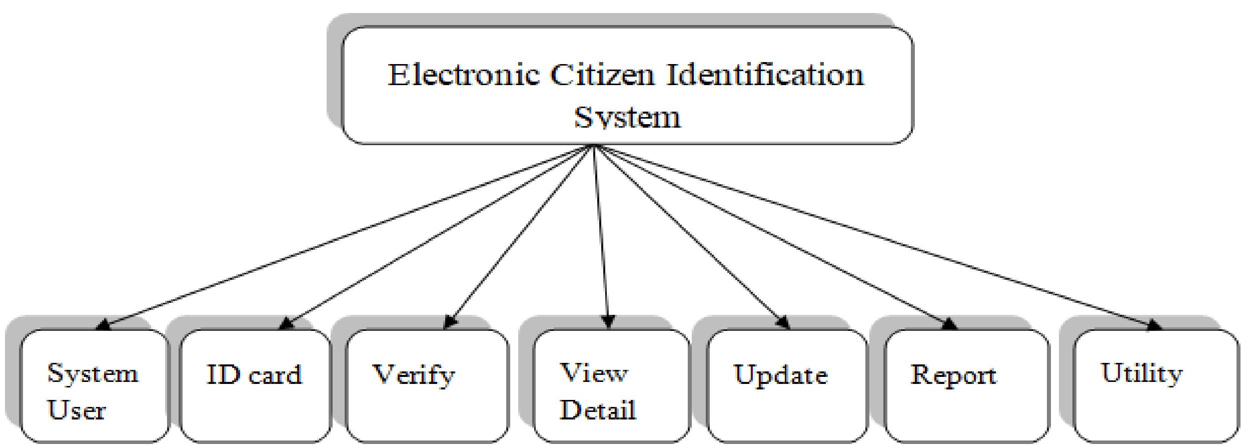

Figure 4. 1 System decomposition

4.2.2 Module description

1) System user

Our system have many user namely user, Citizen, External organization, police officer and admin. Diagram below describe the user and hierarchy. This model deals with the registration or addition and deletion of users of system.

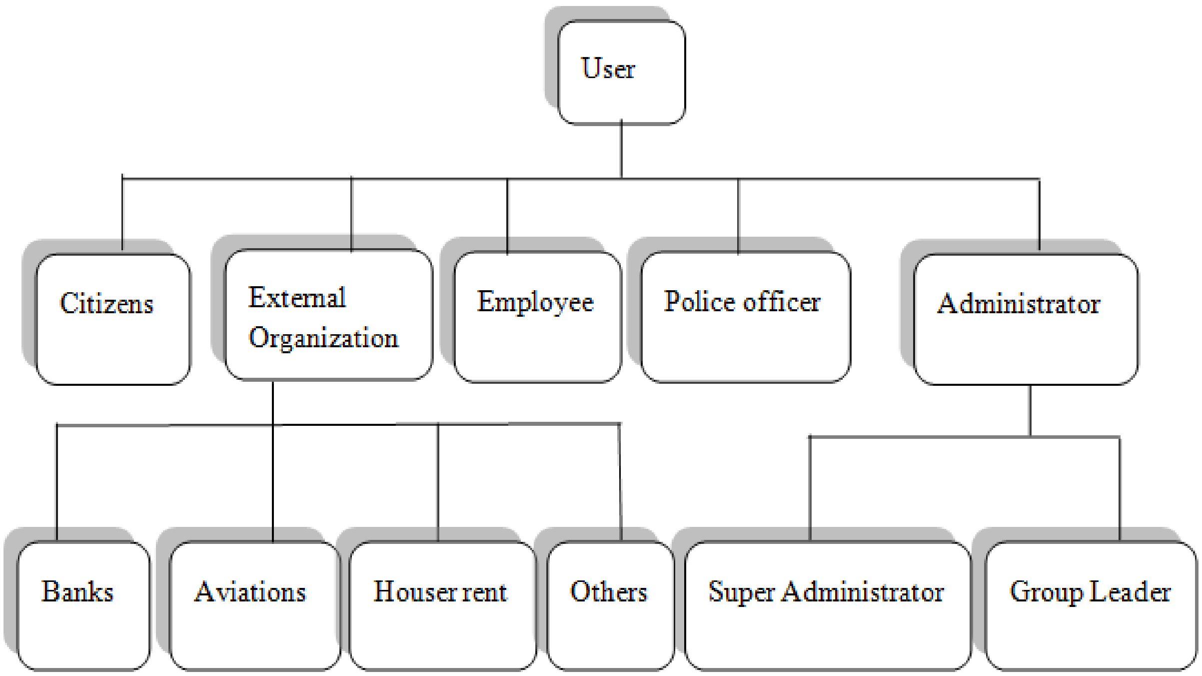

Figure 4. 2 Hierarchy of system user

The module manly contains the following functions.

- **Registering users**

All users should be registered in order to use this system. There are two types of ways to register in the system

- ✓ **Registering as citizen**: - one can be registered as a citizen, so that its data is properly organized and stored in database and available for use. The citizen should fulfill all requirements in order to be registered as citizen. Requirements could be document requirement, data requirement and other requirements. This registration process is done by authorize employees.

- ✓ **Registering as user (create account): -** One can be registered as a user so that it can have access to the system. After the registration process, the user will have its own account.

Provided interface: Registration form

Required interface: System home page

Processing:

- **Register as citizen**
  - ✓ Fill citizen information
    - Full name
    - Date of birth
    - Age
    - Address
    - Gender
    - Place of birth
    - Nationality
    - Job
    - House number
    - Phone number
    - Registration date
    - Blood type
    - Crime status
    - Service status
    - Marital status
    - Education status
  - ✓ Filled Information is verified
  - ✓ Document requirement are uploaded.
  - ✓ Citizen is registered (note that if citizen is applying for registration online, after the registration process is finished, citizen has to go onsite and Employee or authorized body has to approve it).
- **Create account**

- ✓ Choose user type
- ✓ Fill basic information according to the type of user
- ✓ Filled information is verified
- ✓ Upload document requirement
- ✓ Account created and login page displayed. Note that account have to be activated before login by going to **IDCO** (Identification card office).

Table 4. 1 System users detail information

Admins and Employee	External organization	Police officer	Ordinary User
• **ID number** • **First name** • **Last name** • **Position** • **Email** • **Phone number** • **Date of joined** • **Education status**	• ID number • Name • Type • Address • Tin number • Tel phone number • Po.box • Email • Branch(if any)	• ID number • First name • Last name • Position • Police station Branch name • Police station Branch address • Title	• Citizenship ID number • Password • Username • Email

- **Access information**

Citizens information stored on database are protected and available for authorized user. In order to have access to information someone have to login to account and the following access information are required.

- ✓ Username
- ✓ Password

Different users of the system have different access level. For example ordinary user has little access level but admins or employees have higher access level.

Level of access 1

Ordinary users have access level of one. They can view information the same to that of information written on citizen Identification card. In addition to this they can view the following information.

- ✓ They can view and check the validity and legality of Identification cards.
- ✓ They can view crime status of citizen

Level of access 2

Organization has access level of 2.Beside details provided in identification card. They can view information or citizen detail related to their work. Example banks can view whether citizens has bank service ban or not and aviation can view whether citizen has emigration ban.

Level of access 3

Police officers have access level of 3. They have the authority to view most of citizen's information relating to crime status and other detail that police officer need. Also have the authority to modify some citizen information, but those changes have to be approved by authorized employees or admins.

Level of access 4

Employees have access level of 4. Any authorized employee has full access to citizen information. They have full access to the following things.

- ✓ View all citizen information. They can view all citizen information; there is no hidden information to them.
- ✓ Modify citizen information. They can modify citizen information if there is sufficient reason for modification.

Level of access 5

Branch admins (team leader) have access level of 5. They can do everything employees can do. They have access to see report. And they also have access to post and delete notification.

Level of access 6

Super admin has higher level of access. They have full access of the system. The following are exclusively given to them.

- ✓ They can add (approve request for registration) employee
- ✓ Remove employee
- ✓ Modify employee information.

Table 4.2 Access level of system users

Level of access	*User*
1	Ordinary user
2	External organization
3	Police officer
4	Employee
5	Admin (Team leader)
6	Super admin

- **Remove**

Citizens who are no longer eligible are removed from system. Example peoples who are no longer residents or citizens of a country will be removed from system. Employee can remove or add citizens to blacklist.

Employee and Admins who are no longer eligible are removed from the system. This is performed by super admins.

2) ID cards

This module contains all function relating to ID cards. Those functions include preparing new ID card, print ID card and renewing ID card.

Provide Interface: Interface for preparing ID

Required interface: Home page of user

Processing:

- Preparing new ID card:
  - ✓ Fetching citizen data from database.
  - ✓ Generate barcode
  - ✓ Organize necessary information and put on ID card along with the barcode
  - ✓ Put signature on ID card
  - ✓ Store the ID card on database(soft copy)
- Renewing ID card
  - ✓ Fetch ID card
  - ✓ Perform changes if any
  - ✓ Store the ID card on database.
- Printing
  - ✓ Fetch ID card from database
  - ✓ Preview ID card
  - ✓ Print ID card

3) Verify

Contains all functions that deals with validity, legality, expire date of ID cards and citizen information. Every user can verify an ID by login in to their account.

Interface provided: Verification form

Interface required: System home page

Processing:

- Check legality
  - ✓ Read barcode on the ID card or input ID card number
  - ✓ Check if This ID card is in database

- ✓ If not found it is not valid.

- Check validity
    - ✓ Read barcode on the ID card or input ID card number
    - ✓ Check if This ID card is banned or blocked
    - ✓ If it does, it is invalid.
- Check expire date
    - ✓ Read barcode on the ID card or input ID card number
    - ✓ Check expire date of ID card
    - ✓ If ID card is it expired it is invalid.
- Check for service ban
    - ✓ Read barcode on the ID card or input ID card number
    - ✓ Check if flag for service ban has been set
    - ✓ If set service cannot provided

4) View Detail

This module deals with displaying citizen and employee information. Different users need different information. This model provides citizen and employee information based on the user type and authority level.

Interface Provided: Display page

Interface Required: Users home page

Processing:

- View Citizen Detail:- any user can view limited amount of citizen information
    - ✓ Read barcode on the ID card or input ID card number
    - ✓ Search the citizen in database.
    - ✓ If found display necessary information.
- View user (Employee, organization, police officer):- Admin can view users' detail.
    - ✓ Input name or other attribute
    - ✓ Search the user in database.
    - ✓ If found display necessary information.

5) Update

This module provides away to modify citizens or users information.

Interface provided: Form for updating

Interface Required: System home page

Processing:

- Update Citizen Detail: - Authorized Employee or admin can modify or update citizen information.
  - ✓ Input ID number or other attribute
  - ✓ Search the citizen in database.
  - ✓ If found perform the modification.
- Update user (Employee, organization, police officer):- Admin can modify users' information.
  - ✓ Input name or other attribute
  - ✓ Search the user in database.
  - ✓ If found perform the modification

6) Report

This module generated report to admin .It provides different report to team leaders and super admin.

Interface provided: Report page

Required Interface: System home page

Processing:

- Generate report about number of citizen: - Module provides report about number of citizen by categorizing them by age, sex and other information. This report is provided to employees and admin.
  - ✓ Module fetch citizen data from database
  - ✓ Process data
  - ✓ Provide report when asked
- Generate report about number of user: - Module provides report about number of user by categorizing them by type other information. This report is provided to super admins.

- ✓ Module fetch user data from database
- ✓ Process data
- ✓ Provide report when asked

- Generate report about actions and changes made: - Module Provide report about the registrations made, modifications made, ID prepared. This report gives details who perform these action when did those action performed. This report is provided to team leader and admins.
  - ✓ Employee or admin make any change to system.
  - ✓ Record detail of these changes
  - ✓ Provide report detail when asked

7) Utility

This module contain utility functions such as post notifications, display newsfeeds and setting user account.

Interface provided: Utility interface

Interface required: System home page

Processing:

- Posting notification:- admin and team leader can post notification
- Deleting notification: - team leader can delete its own post while super admin can delete nay post.
- Seeing notification: - Users can view both public post and notification that are specific to them.
- Setting user account:- Users can set their own account

4.3 Architecture of the system

4.3.1 Architectural style and pattern

The architectural style and pattern for this system are chosen in way that makes the development process easier and in the way that satisfy requirements.

Architectural style

- **Language Based – Object Oriented: -** This architectural style makes the development and maintenance process of our system easier by dividing the whole system in to manageable and reusable and components, modules and objects.
- **Layered** – **Client Server: -** This architectural style divides many operations and function of our system and assigns them to be processed on different computer and distribute the work load to those computers.

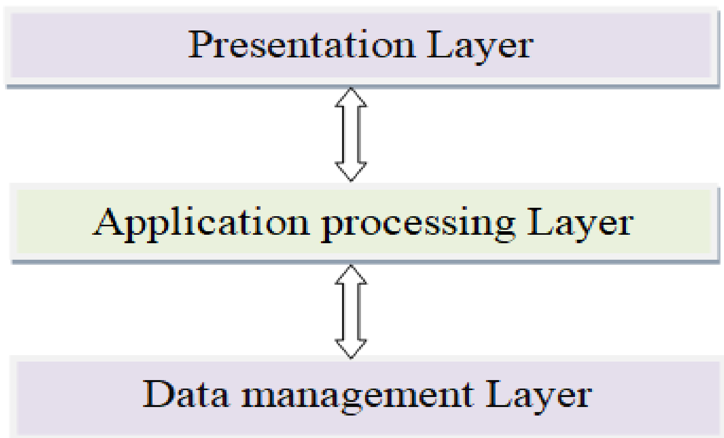

Figure 4. 3 Client server architecture

Architectural pattern

MVC (Model-view-controller): - This architectural pattern separates an application into three main logical component. Our system is separated into three logical components as follows.

Model: - Data in MongoDB database

Controller: - Back end of system build by NodeJs

View: - Front end user interface build by HTML, CSS, React JS

4.3.2 Component Diagram

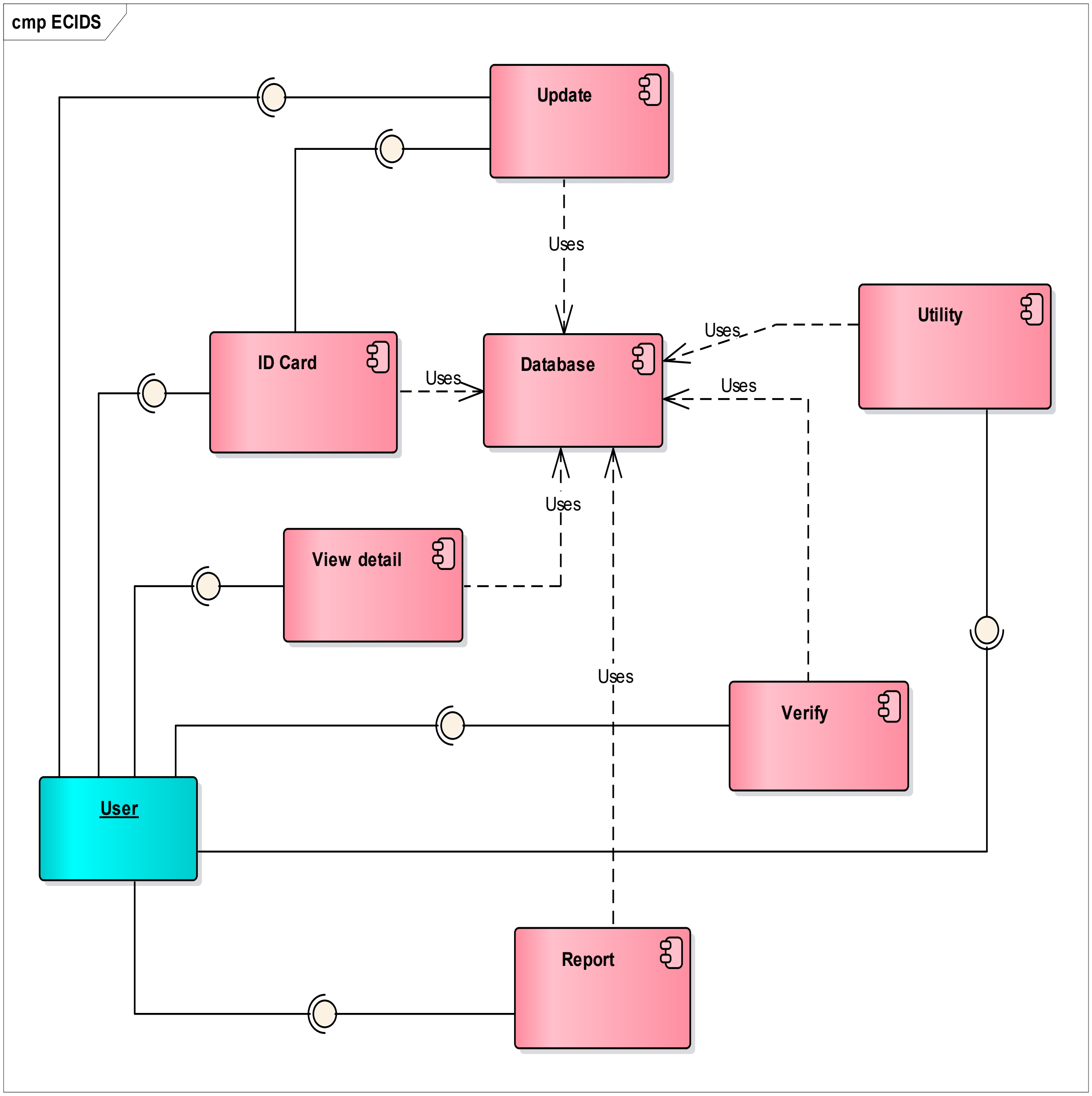

Figure 4. 4 Component diagram of ECIDS

4.3.3 Deployment Diagram

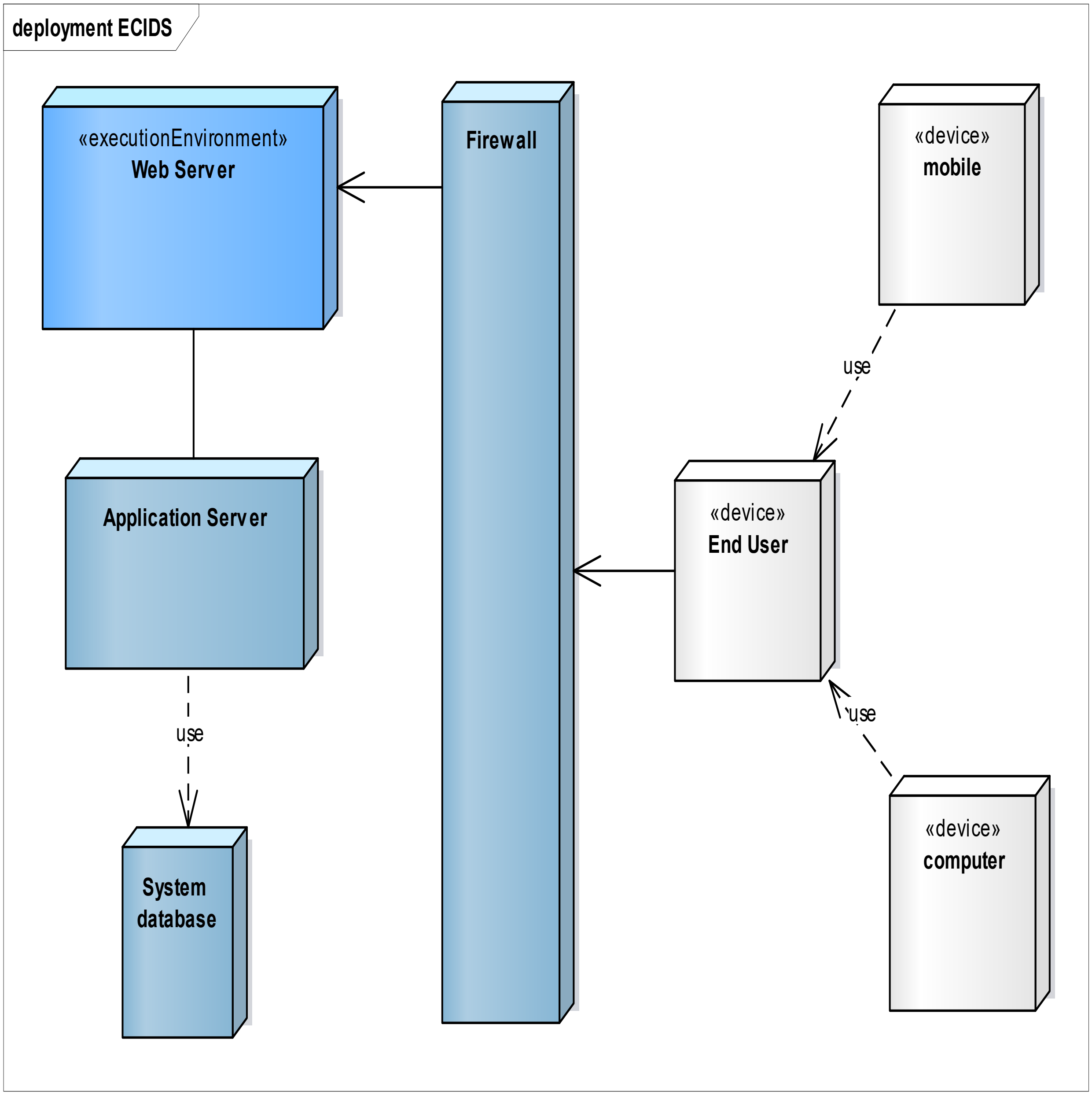

Figure 4. 5 Deployment diagram of EID

4.4 Database Design

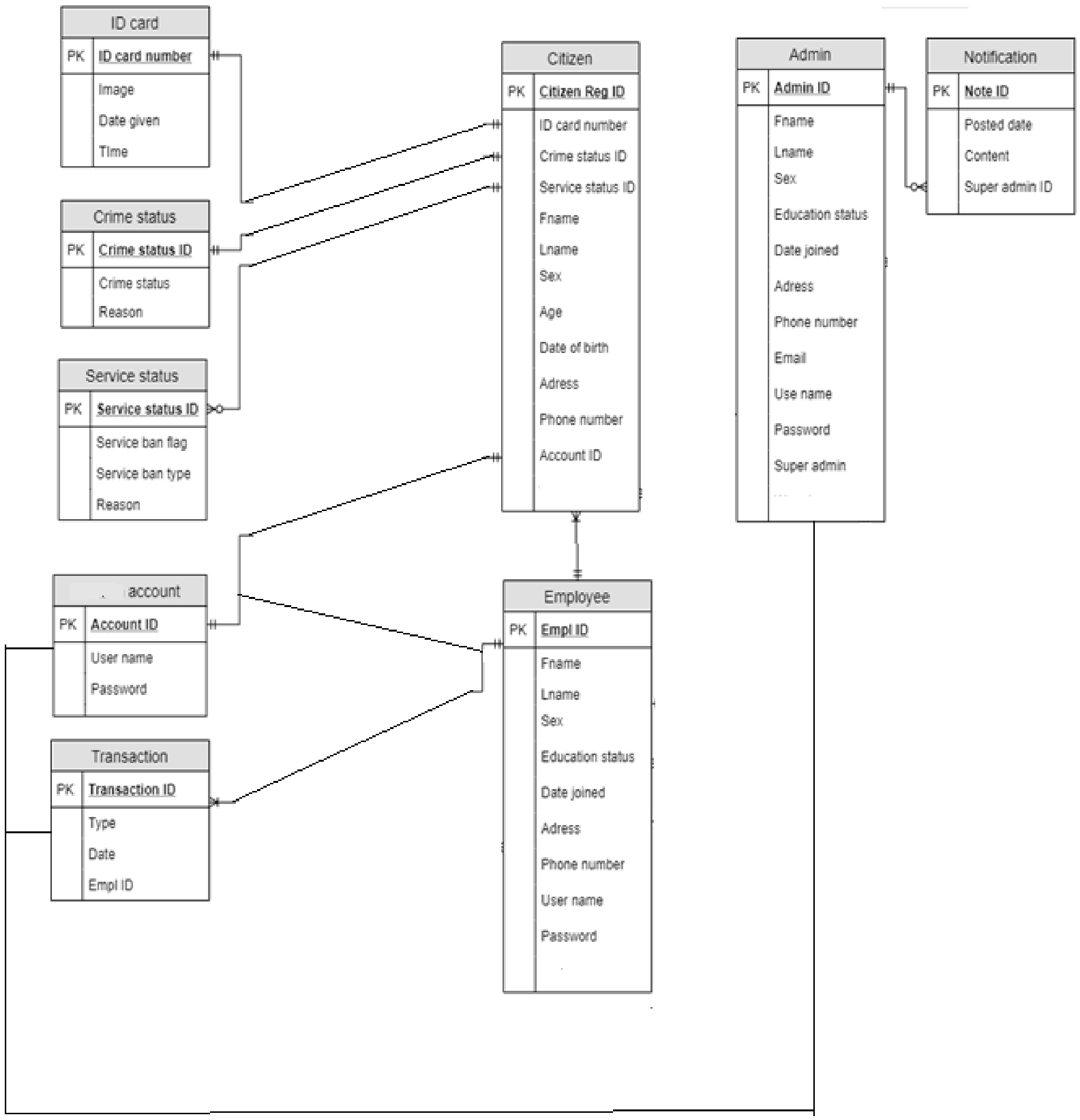

Figure 4. 6 Entity relationship diagram for ECIDS

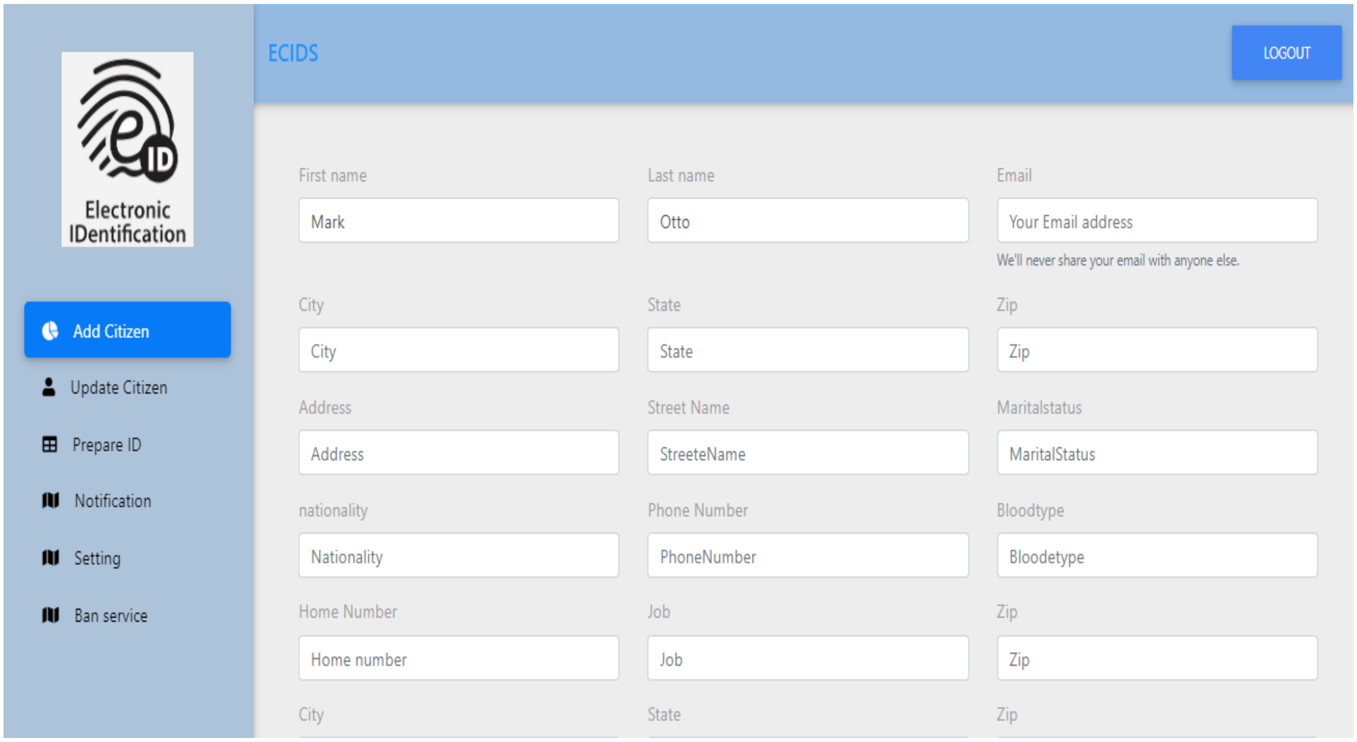

Figure 4. 7 UI design for registration form

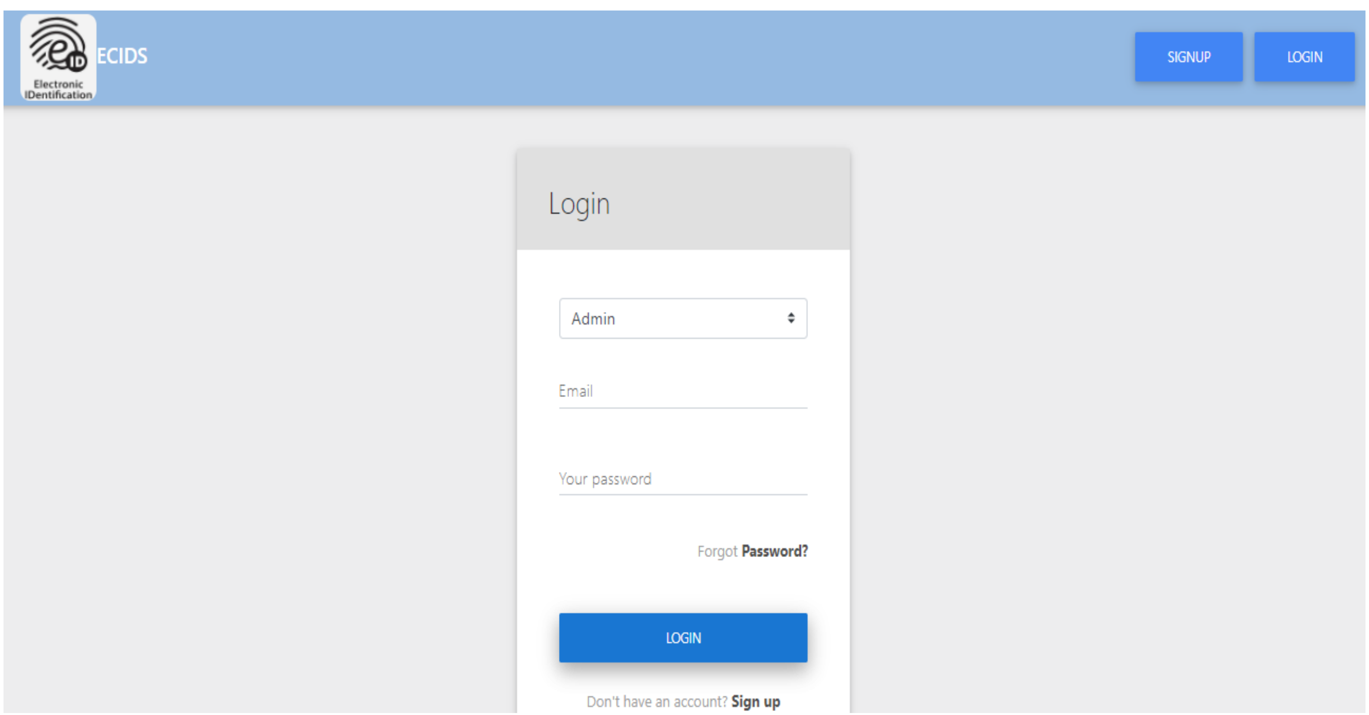

Figure 4. 8 UI design for login page

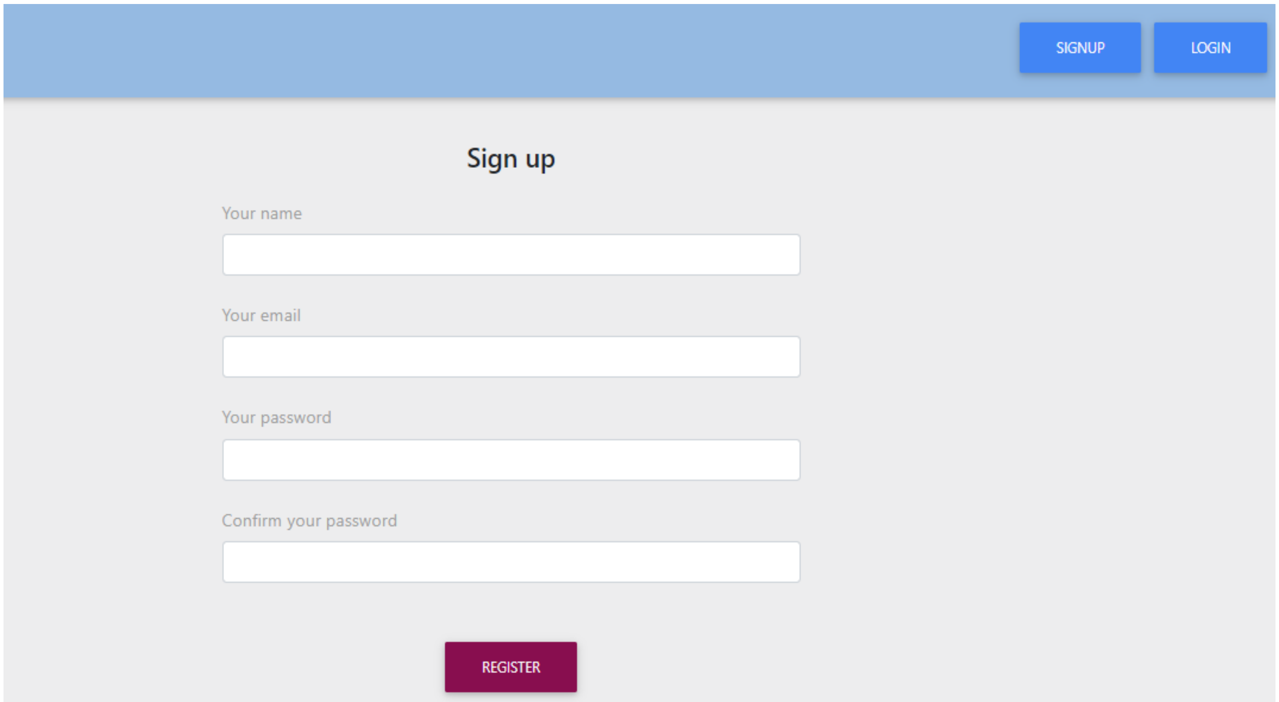

Figure 4. 9 ***UI design for signup page***

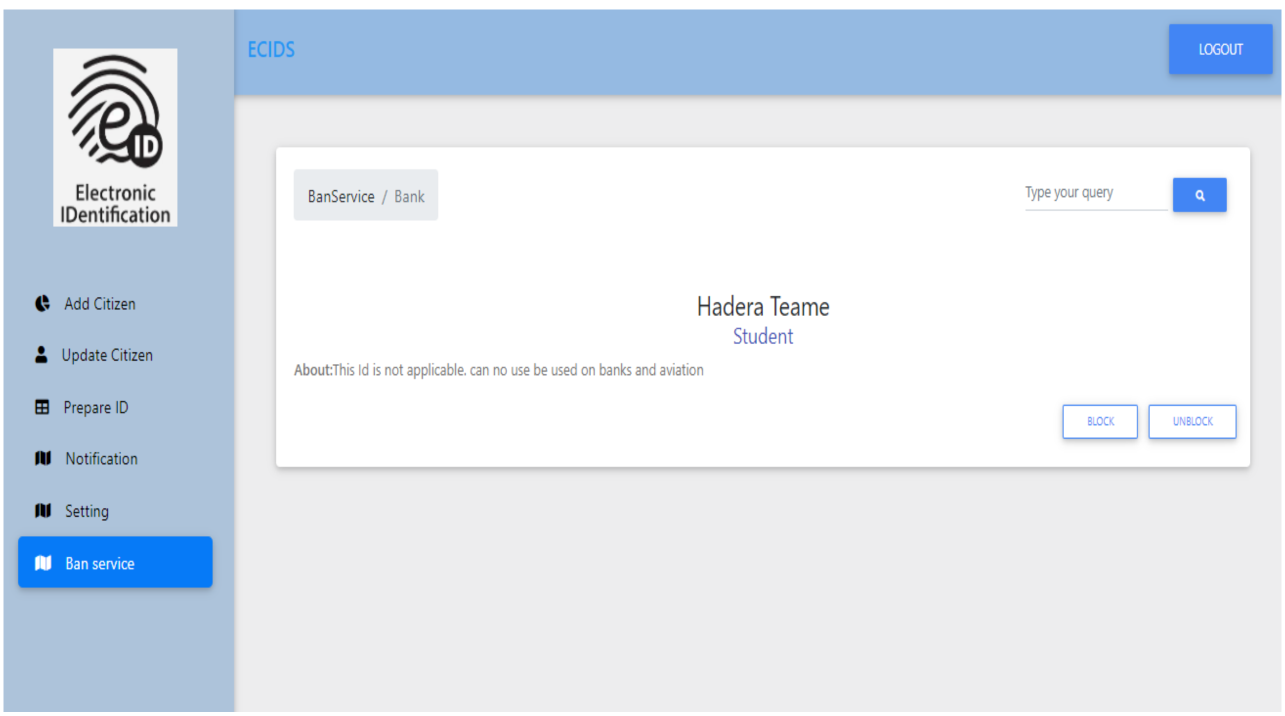

Figure 4. 10 UI design for banning citizen from service

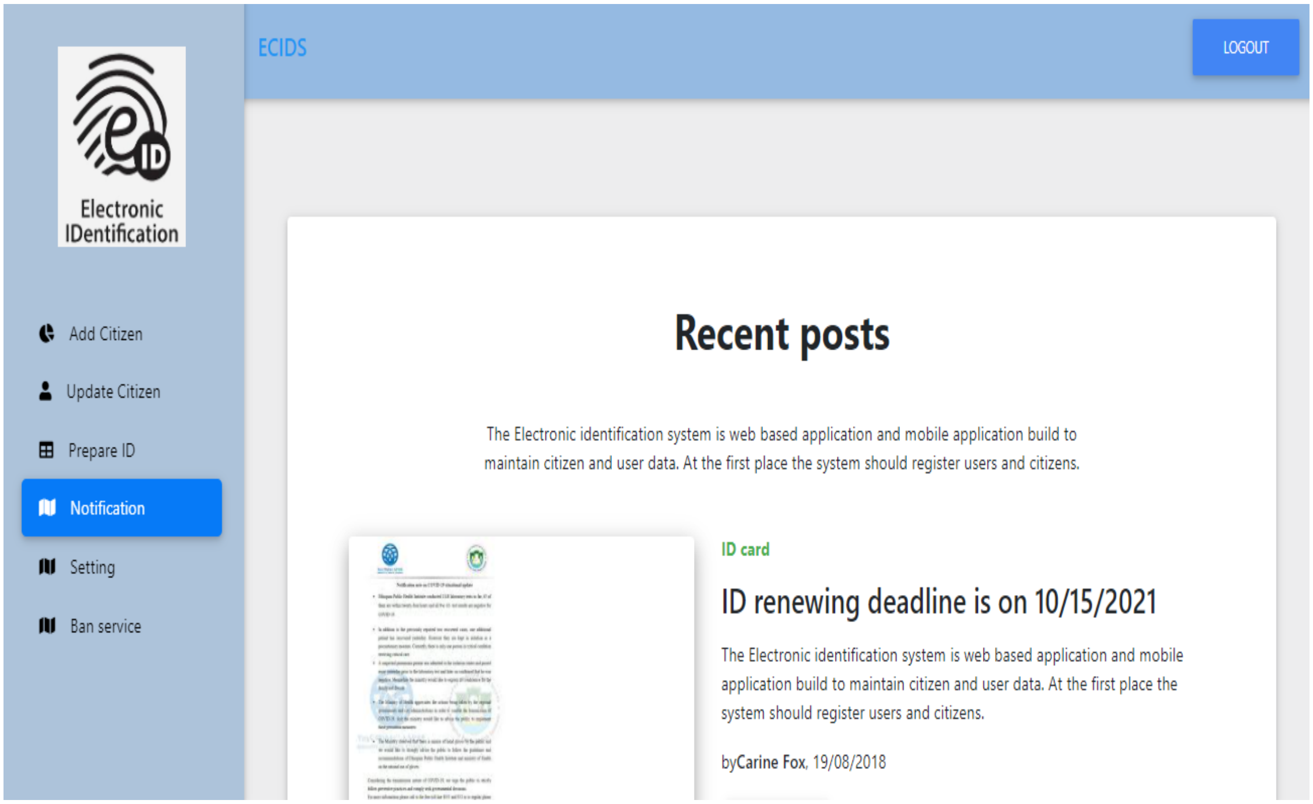

Figure 4. 11 UI design for notification posted page

CHAPTER FIVE: Testing and Implementation

5.1 Implementation

This section discuss about the Implementation phase of our project which is part of our software development life cycle in which the physical system specifications are converted in to a working and software system or solution. The implementation process was divided in to parts, namely front-end and back-end implementation. In the implementation process we use a lot of tools, language and frame-works both at the back-end and front-end implementation.

5.1.1 Software tools, language and frame work

- **React JS: -** Java script library used as to develop our web base application front end. We choose this frame because it make user interface responsive and attractive.
- **Redux:** - is JavaScript library used for state management of component. Mostly used with in other JavaScript library like react and angular.
- **Visual Studio code: -** Text editor used for editing and running our code. This text editor has terminal or command promote integrated with it which makes our work easy both at front-end and back-end implementation.
- **GitHub: -** version control system which helps as manage changes made to our code.
- **Node JS: -** Java scrip frame work used for developing the back end of web based application system.
- **Mongo Db: -** No-SQL and n-relational database that minimize the time spent to query or fetch data.
- **Material design bootstrap: -** this library combined Material design and bootstrap. It is an open source toolkit based on Bootstrap for developing Material Design apps with HTML, CSS, and JS. Quickly prototype your web application.

5.1.2 Software Implementation classification

Front-end

Front-end implementation is the development of user interface that is going to be render and viewed by the end-user on the client side browser.

Back-end

Back-end implementation is the development of the core part of the system that do main computation, transaction and functionality of the system and that is going to be deployed on server, waiting for request from the client application.

5.1.3 Source code

Front-end login form

```
import React , {Component}from "react";
import { MDBContainer, MDBRow, MDBCol, MDBBtn, MDBCard, MDBCardBody,
MDBInput } from 'mdbreact';
import '../index.css';
import PropTypes from "prop-types";
import { connect } from "react-redux";
import { loginUser } from "../actions/authActions";
import classnames from "classnames";

class Login extends Component {
  constructor() {
    super();
    this.state = {
      email: "",
      password: "",
      errors: {}
    };
  }

  componentDidMount() {
```

```
    // If logged in and user navigates to Login page, should redirect them to dashboard
    if (this.props.auth.isAuthenticated) {
      this.props.history.push("/dashboard");
    }}
  componentWillReceiveProps(nextProps) {
    if (nextProps.auth.isAuthenticated) {
      this.props.history.push("/dashboard"); // push user to dashboard when they login
    }
if (nextProps.errors) {
      this.setState({
        errors: nextProps.errors
      });}}
onChange = e => {
    this.setState({ [e.target.id]: e.target.value });
  };
onSubmit = e => {
    e.preventDefault();
const userData = {
      email: this.state.email,
      password: this.state.password
    };
this.props.loginUser(userData);
  };
render() {
    const { errors } = this.state;
return (
      <MDBContainer className="form-login" >
        <MDBRow>
          <MDBCol md="2" lg="4">
          </MDBCol>
          <MDBCol md="8" lg="4">
            <MDBCard>
              <div className="header pt-3 grey lighten-2">
```

```
        <MDBRow className="d-flex justify-content-start">
          <h3 className="deep-grey-text mt-3 mb-4 pb-1 mx-5">
            <center>Login</center>
          </h3>
        </MDBRow>
      </div>
      <MDBCardBody className="mx-4 mt-4">
      <div>
        <select className="browser-default custom-select">
          <option value="1">Admin</option>
          <option value="2">Employee</option>
          <option value="3">User</option>
        </select>
    </div>
    <form noValidate onSubmit={this.onSubmit}>
        <MDBInput label="Email" group  validate
          onChange={this.onChange}
          value={this.state.email}
          error={errors.email}
          id="email"
          type="email"
          className={classnames("", {
            invalid: errors.email || errors.emailnotfound
          })}/>
        <span className="red-text">
          {errors.email}
          {errors.emailnotfound}
        </span>

        <MDBInput label="Your password" group validate
          containerClass="mb-0"

          onChange={this.onChange}
```

```
  value={this.state.password}
  error={errors.password}
  id="password"
  type="password"

  className={classnames("", {
    invalid: errors.password || errors.passwordincorrect
  })}
/>
<span className="red-text">
  {errors.password}
  {errors.passwordincorrect}
</span>
<p className="font-small grey-text d-flex justify-content-end">
  Forgot
  <a
    href="#!"
    className="dark-grey-text font-weight-bold ml-1"
  >
    Password?
  </a>
</p>
<div className="text-center mb-4 mt-5">
  <MDBBtn
    color="blue"
    type="submit"
    className="btn-block z-depth-2 log-btn"
  >
    Login
  </MDBBtn>
</div>
</form>
<p className="font-small grey-text d-flex justify-content-center">
```

```
              Don't have an account?
              <a
                href="#!"
                className="dark-grey-text font-weight-bold ml-1"
              >
                Sign up
              </a>
            </p>
          </MDBCardBody>
        </MDBCard>
      </MDBCol>
    </MDBRow>
  </MDBContainer>
 );
 };
}
Login.propTypes = {
  loginUser: PropTypes.func.isRequired,
  auth: PropTypes.object.isRequired,
  errors: PropTypes.object.isRequired
};
const mapStateToProps = state => ({
  auth: state.auth,
  errors: state.errors
});
```

Front-end employee dashboard

```
import React, { Component } from 'react';
import Routes from './Routes';
import TopNavigation from './TopNavigation.js';
import SideNavigation from './SideNavigation.js';
import Footer from './Footer';
```

```
import '../index.css';
import { MDBContainer, MDBRow, MDBCol, MDBBtn } from 'mdbreact';

import PropTypes from "prop-types";
import { connect } from "react-redux";
import { logoutUser } from "../actions/authActions";

class DashBoard extends Component {
  onLogoutClick = e => {
    e.preventDefault();
    this.props.logoutUser();
  };
render() {
    const { user } = this.props.auth;

return (
<div className="flexible-content">

        <TopNavigation onLClick={this.onLogoutClick}/>
        <SideNavigation type="employee"/>
        <main id="content" className="p-5">
          <Routes type="employee"/>
        </main>
      </div>
); }}
DashBoard.propTypes = {
  logoutUser: PropTypes.func.isRequired,
  auth: PropTypes.object.isRequired
};
const mapStateToProps = state => ({
  auth: state.auth
});
export default connect(
```

```
  mapStateToProps,
  { logoutUser }
)(DashBoard);
```

Back-end for sign-up and login

```
const express = require("express");
const router = express.Router();
const bcrypt = require("bcryptjs");
const jwt = require("jsonwebtoken");
const keys = require("../../config/keys");
// Load input validation
const validateRegisterInput = require("../../validation/register");
const validateLoginInput = require("../../validation/login");
// Load User model
const User = require("../../models/User.js");
// @route POST api/users/register
// @desc Register user
// @access Public
router.post("/register", (req, res) => {
    // Form validation
  const { errors, isValid } = validateRegisterInput(req.body);
  // Check validation
    if (!isValid) {
      return res.status(400).json(errors);
    }
  User.findOne({ email: req.body.email }).then(user => {
      if (user) {
        return res.status(400).json({ email: "Email already exists" });
      } else {
        const newUser = new User({
          name: req.body.name,
          email: req.body.email,
          password: req.body.password
```

```
    });
 // Hash password before saving in database
    bcrypt.genSalt(10, (err, salt) => {
     bcrypt.hash(newUser.password, salt, (err, hash) => {
      if (err) throw err;
      newUser.password = hash;
      newUser
       .save()
       .then(user => res.json(user))
       .catch(err => console.log(err));
     });
    });
   }
  });
 });
// @route POST api/users/login
// @desc Login user and return JWT token
// @access Public
router.post("/login", (req, res) => {
  // Form validation
 const { errors, isValid } = validateLoginInput(req.body);
 // Check validation
  if (!isValid) {
   return res.status(400).json(errors);
  }
 const email = req.body.email;
  const password = req.body.password;
 // Find user by email
  User.findOne({ email }).then(user => {
   // Check if user exists
   if (!user) {
    return res.status(404).json({ emailnotfound: "Email not found" });
   }
```

```
// Check password
    bcrypt.compare(password, user.password).then(isMatch => {
      if (isMatch) {
        // User matched
        // Create JWT Payload
        const payload = {
          id: user.id,
          name: user.name
        };
// Sign token
        jwt.sign(
          payload,
          keys.secretOrKey,
          {
            expiresIn: 31556926 // 1 year in seconds
          },
          (err, token) => {
            res.json({
              success: true,
              token: "Bearer " + token
            });
          }
        );
      } else {
        return res
          .status(400)
          .json({ passwordincorrect: "Password incorrect" });
      }
    });
  });
});
module.exports = router;
```

5.2 Testing

Testing is one phase in Software development life cycle. Any implemented system or software has to pass through this phase in order to assure software quality. This is achieved by checking whether the system is defect free or not and whether the system meet the specified requirement or. We implemented various techniques and step of testing in order to check whether our system is error free and valid. Some of the methods and techniques we use are mentioned below.

5.2.1 Testing phases

Fundamental functional Testing: Parallel with the implementation process also after the implementation process we made frequent basic testing whether every component and every function is functioning well or not. This Testing method help us to detect and debug many defects at early stage before using or deploying the system.

Code Review: we reviewed the entire code both the front-end and back-end. This help as to update the code according to the test result, so as modify the code, find bug, restructuring the code after testing

Unit Testing: we examine every unit of the system with the help tool. We developed test cases for independent and individual unites (component. Methods, classes, functions, and interfaces) to check whether they are valid using some tool.

Integration Testing: After we made unit testing to individual units, we integrate the individual units together and test them as whole and check their integrity and their interface where two units or components communicate.

System Testing: This testing phase is up to checking the validity of the system as whole. After performing all the above test we evaluate the whole system as per the specified requirement (functional and non-functional requirement).

User Acceptance Testing: Due to the time limitation we don't perform the user acceptance testing and we are unable to get feedback from the end-user of the system. We will deploy the system and perform user acceptance testing with help of users and after accepting feedback from users we will update the system on the next version or release as per the feedback.

5.2.2 Test case execution

Generally we prepare test-case and run the test case and compare the result with the expected result. So as to examine and evaluate the gap between the result and the expected result. The total step is summarized on the following table.

- For each executed test, this document contains:
    - Test identification;
    - Test title;
    - Test decision;
    - A comment containing additional information or problems encountered during execution and differences with the test procedure.
    - Also the result of that comes as the test case executed
    - Remark are going the following comments
        - If the actual outcome is 100% equal to the expected outcome remark going to be = ***PASS;***
        - If the actual outcome is 50%-90% similar to the expected outcome remark going to be = ***MODERATE;***
        - If the actual outcome is <50% similar to the expected outcome or totally different remark going to be = ***FAIL;***

Detailed Test case execution

Table 5. 1: Table of test case and the test result after testing

Test description	Scenario	Test step	Expected out come	Actual Outcome	Remark
• ID:001 • Name: Login • Precondition: User has an	Correct password and	Input Correct password and Correct user	Homepage of the user	The system properly authenticates	PASS

account	Correct user name	name to the system		the user and loads proper homepage for the user	
	Incorrect user name and password	Input Incorrect user name and password to the system	Display Incorrect user name and password output	The system properly Display the expected output	PASS
	Correct password but incorrect user name	Input Correct password but incorrect user name to the system	Display Correct password but incorrect user name output	The system properly Display the expected output	PASS
	Correct username but incorrect password	Input Correct username but incorrect password to the system	Display Correct username but incorrect password output	The system properly Display the expected output	PASS
• ID:002 • Name: Register • Precondition: No precondition	Register Citizen	Fill the necessary input and submit register	Display success full registered and add the input data to the data base	Successfully add input data to database but it does not display notification	MODERATE

	Register Employee	Fill the necessary input and submit register	Display success full registered and add the input data to the data base	Successfully add input data to database but it does not display notification	PASS
	Register User	Fill the necessary input and submit register	Display success full registered and add the input data to the data base	Successfully add input data to database but it does not display notification	PASS
	Try to register with invalid input	Fill invalid input to all input in the form	Display there is invalid input in your form and prevent you from registering	It does not display any massage and allowed to be registered	PASS
• ID:003 • Name: Generate bar code • Precondition: No precondition	Generate bar code	Press the generate barcode	Produce and display barcode with unique ID	Successfully produce and display barcode	MODERATE
• ID:004 • Name: Generate ID number • Precondition: No precondition	Generate ID string	Ask the system to generate Unique ID String	Yields Unique ID string	Successfully Yields Unique ID string	PASS

• ID:005 • Name: Prepare ID Card • Precondition: No precondition	Prepare ID cards	Ask the system to Prepare ID cards	Display ID card and store it in database	Successfully display Id card and stored it	PASS
• ID:006 • Name: Modify • Precondition: No precondition	Modify Employee	Fill the necessary input and submit modify	Display success fully modified and Update database	Successfully update data on database	PASS
	Modify User	Fill the necessary input and submit modify	Display success fully modified and Update database	Successfully update data on database	MODERATE
	Modify Citizen	Fill the necessary input and submit modify	Display success fully modified and Update database	Successfully update data on database	PASS
	Try to modify with invalid input	Fill invalid input to all input in the form	Display there is invalid input in your form and prevent you from modifying	Allowed data to be updated. Successfully update data on database	PASS

• ID:007 • Name: Customize Account • Precondition: No precondition	Customize or set account with valid input	Fill valid input to all input in the form	Change account information And notify	Account information changed and notify	PASS
	Customize or set account with invalid input	Fill invalid input to all input in the form	Do not Change account information And notify error	Do not Change account information And notify error	PASS
• ID:008 • Name: Display • Precondition: No precondition	Display Citizen data	Search and press Display information button	Citizen information Displayed	Citizen information Displayed	PASS
• ID:009 • Name: Verify • Precondition: No precondition	Verify valid citizen	Input ID of valid citizen or citizen out of database list Search and press verify citizen information button	Display Citizen is valid	Displayed Citizen is valid	PASS
	Verify invalid citizen	Input ID of invalid citizen or citizen out of database list	Display Citizen is invalid	Display Citizen is invalid	PASS

		Search and press verify citizen information button			
• ID:010 • Name: Block ID • Precondition: No precondition	Block ID	Search user and press the block Id button	Citizen Id must be blocked	Citizen is blocked	PASS
• ID:011 • Name: Post Notification • Precondition: No precondition	Post notification	Input the necessary data to be posted and press post or submit	Post are stored to database and posted	Notifications posted and stored on Database	PASS
• ID:012 • Name: Search • Precondition: No precondition	Search	Input data and submit search	Display the needed result	Needed data displayed	PASS

CHAPTER SIX: Conclusion and future work

6.1. Future work

Our system implement almost all core functionality. But some functionalities relatively having low priority are not included in this release. In the future we or our team-mate are going to update the system by implementing not implemented functional and non-functional requirement. Those requirements was not included in this release due limitations mentioned above such as time limitation and covid-19 pandemic disease.

In the future release, we are going to come up with the mobile application for this system in addition to the web based application. In addition we are going to extend the system by adding functionality to the system as per the feedback of the client and changing in requirement. We are also going to extend the scope of the project in order to make it applicable at national level.

6.2. Conclusion

Citizen Identification is becoming big issue this days because the identification process key in our day to day life and it is need for many real-life applications and services. Because of this the Identification process should be digitalize and automated in order to improve the traditional and current identification system.

We believe, we improve the current identification process. By coming up with high qualified software system solution to improve the problem that arises from the current systems. And we do not believed that this is just the end, the identification process needs further modification and improvement in the future.

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APPENDIX 1 – Questioner paper

Questioner paper For Citizen

- ይሄ መጠይቅ ከመታወኪያ ተጠቃሚዎች መረጃ ለመሰብሰብ የተዘጋጀ ነው ፤ እና እባክዎ ያዩት፣የሚያውቁት፣የሰሙት እና ባጠቃላይ ያሎት መረጃ በሃላፊነት እንዲሞሉ በትህትና እንጠይቃለን፡፡
- ከታች ላሉትጢያቄዎች ሁለት ኣይነት መልስ ኣላቸው፡፡ ኣንደኛ ከተሰጡት ምርጫዎች ኣንዱ በማጥቆር መምረጥ ሲሆን ሁለተኛው በተሰጡት ክፍት ቦታዎች የምታቁት ሃሳብ መሙላት ነው ፡፡

ክፍል 1 ፡ የሚከተሉት ጢያቄዎች ኣገልግሎት እየሰጠ በነበረው የወረቀት ፋይል መሰረት ያደረገ የመታወቂአ ስርኣት ላይ ያተኮሩ ናቸው፡፡

1. ኣዲስ መታወቅያ ለማውጣት ስንት ቀን ተመላልሰዋል ?

☐አንድ ቀን ☐ ሁለት ቀን ☐ ሶስት ቀን ☐ከ ሶስት ቀን በላይ

2. ኣዲስ መታወቅያ ለማውጣት መጥተህ በቀን ስንት ሰአት አባክነዋል ?

☐አንድ ሰአት ☐ሁለት ሰአት ☐ግማሽ ቀን ☐ ሙሉ ቀን

3. የ ድሮ የመታወቅያ አሰጣጥ ስርኣት ከ ፋይል ጋር የተያያዘ ነው፡፡ ፋይልዎ ወይም ፋይልዎ ዉስጥ ያሉ መረጃዎች ጠፍትዎት ያውቃል ?

☐ያውቃል ☐አያውቅም

4. ፋይል የጠፋበት ሰው ያውቃሉ ወይም ፋይል ስለ ጠፋበት ሰው ሰምተው ያውቃሉ ?

☐አዎ ☐አይ

5. በ ድሮ መታወቅያ አሰጣጥ ላይ ችግር አለ ? ☐አዎ ☐የለም
   a. መልስህ ኣዎ ከሆነ ምን ምን ችግሮች አሉ?

1. ____
2. ____

6. በ ድሮ መታወቅያ አሰጣጥ ላይ ኣገልግሎት ለማግኘት ሂደው እንግልት ደርሶቦት ያቃል ?

☐አዎ ☐የለም

   a. መልስህ ኣዎ ከሆነ ምን ኣይነት እንግልት ደርሶቦት ?

1. ____
2. ____

7. በ ድሮ የመታወቅያ አሰጣጥ ደስተኛ ነበሩ? ምን ያህል?

☐ በጣም ደስተኛ ነኝ ☐ደስተኛ ነኝ

☐ ለክፉ አይሰጥም ☐ደስተኛ አይደለሁም

☐ በጣም ጠልችዋሎህ

ክፍል 2 ፡ አዲስ የመታወቅያ አሰጣጥ ዘዴ በ አዲስ አበባ ተግባራዊ ሊሆን ዝግጅት ላይ ነው፡፡የሚከተሉት ጢያቄዎች በዚ የመታወቂአ አሰጣጥ ስርኣት ላይ ያተኮሩ ናቸው፡፡

1. ኣዲስ የመታወቅያ አሰጣጥ ዘዴ ተግባራዊ ሊሆን ዝግጅት ላይ ነው ስለዚህ ነገር ምያቁት ነገር አለ?

☐አለ ☐የለም

a. መልስዎ አዎ ከሆነ ምን ምን መረጃ አሎት?

1. ____________________

2. ____________________

8. እስከ ቅርብ ጊዜ አገልግሎት ላያ በነበረ እና ኣሁን ተግባራዊ ሊሆን ዝግጅት ላይ ካለው ስርኣት የሚለይ በ ኮምፒተር የታገዘ የመታወቂያ ስርኣት እየሰራን ነው ይህ አገልግሎት ምን ምን እንዲያሚላ ትፈልጋለህ?

1. ____________________

2. ____________________

Questioner paper For Organization

- ይሄ መጠይቅ ኣገልግሎት ለመስጠት የዜጎች መታወቂያ ከ ሚጠይቁ ድርጅት መረጃ ለመሰብሰብ የተዘጋጀ ነው ፤ እና እባክዎ ያዩት፣የሚያውቁት፣የሰሙት እና ባጠቃላይ ያሎት መረጃ በሃላፊነት እንዲሞሉ በትህትና እንጠይቃለን፡፡
- ከታች ላሉትጢያቄዎች ሁለት ኣይነት መልስ ኣላቸው፡፡ ኣንደኛ ከተሰጡት ምርቻዎች ኣንዱ መምረጥ ሲሆን ሁለተኛው በተሰጡት ክፍት ቦታዎች የምታቁት ሃሳብ መሙላት ነው ፡፡

1. አገልግሎት ፈልገው የሚመጡ ተገልጋዮችን መታወቂያ ትጠይቃላችሁ ? ☐አዎ ☐አይ
   1. መልስዎ ኣዎ ከሆነ ፤ አስፈላጊነቱ ለምን ይሆን?
      1. ____________________
      2. ____________________
   1. መታወቂያው ውስጥ ካሉ መረጃወች መካከል ለናንተ የሚጠቅሟችሁ የትኞቹ ናቸው?
      1. ____________________
      2. ____________________

2. ህጋዊ ያልሆነ ወይም ተቀባይነት የሌለው መታወቂያ በመያዝ የማጭበርበር ሙከራ የሚያደርግ ተገልጋይ ገጥሟችሁ ያውቃል?

☐ያውቃል ☐አያውቅም ☐ማወቅ የምንችልበት መንገድ የለም

1. ኣጋጥሞት የሚያውቅ ከሆነ ፤ ምን የማጭበርበ ኣይነት ኣጋጠምዎ ?
    1. ____
    2. ____

3. አንዳንድ ተገልጋዮች ህጋዊ ያልሆነ መታወቂያ ይዘው የሚቀርቡበት ሁኔታ ሊፈጠር ይችላል፡፡ የመታወቂያ ህጋዊነት ማረጋገጫ ስልቶች ኣላችሁ?

☐ኣዎ ☐ኣይ

- ስልቶች ካላችሁ ፤ ምን ምንድ ናቸው?
    1. ____
    2. ____
  - እነዚህ የምትጠቀሟቸው ስልቶች ወይም መንገዶች አስተማማኝ እና ውጤታማ ናቸው ብላችሁ ታስባላችሁ ?

    ☐ኣዎ ☐ኣይ

  እንዴት? ለመልስኦ ማብራሪያ ይስጡ፡
    - i. ____
    - ii. ____

4. ኣብዛኛው ጊዜ ህጋዊ ባልሆነ መታወቂያ ወንጀል ይሰራል፣ የእርስዎ እይታስ ምንድነው?

☐የተሰራ ወንጀል አውቃለሁ ☐ወንጀል አኔን አጋጥሞኝ ያውቃል

☐የተሰራ ወንጀል በወሬ ደረጃ ሰምቸ አውቃለሁ ☐ኣላውቅም

- ምንድነው? እንዴት? ለመልስዎ ማብራሪያ ይስጡ
  - ____
  - ____

5. በጊዜው ያሉ ችግሮችን ለመቅረፍ ኤሌክትሮኒክስ ወይም ቾምፐዩተራይዝድ የሆነ ስይስተም እየሰራን ነው፣ ስይስተሙ ምን ምን ነገሮችን ቢያካትት ውቴታማ ያደርገናል ብላችሁ ታስባላችሁ(ከ ድርጅዎ እይታ እና ፍላጎት ኣንፃር)?
    1. ____
    2. ____

Questioner paper For Employee

- ይሄ መጠይቅ ከመታወቂያ ሰጪ ድርጅት ሰራተኞች መረጃ ለመሰብሰብ የተዘጋጀ ነው ፤ እና እባክዎ ያዩት፣ የሚያውቁት፣የሰሙት እና ባጠቃላይ ያሎት መረጃ በሃላፊነት እንዲሞሉ በትህትና እንጠይቃለን፡፡
- ከታች ላሉትጢያቄዎች ሁለት አይነት መልስ አላቸው፡፡ አንደኛ ከተሰጡት ምርጫዎች አንዱ መምረጥ ሲሆን ሁለተኛው በተሰጡት ክፍት ቦታዎች የምታቁት ሃሳብ መሙላት ነው ፡፡

ክፍል 1 ፡ የሚከተሉት ጢያቄዎች አገልግሎት እየሰጠ በነበረው የወረቀት ፋይል መሰረት ያደረገ የመታወቂአ ስርአት ላይ ያተኮሩ ናቸው፡፡

1. ከደንበኞች ወይም ከተገልጋዮች ቅሬታ ወይም መጥፎ አስተያየት ተሰጥታችሁ ወይም አጋጥማችሁ ያቃል?

☐አዎ ☐አይ

a. መልስዎ አዎ ከሆነ ምን ቅሬታ ተሰጥታችሁ ያቃል?

1. ______________________________

2. ______________________________

2. በስራችሁ ላይ ችግር አጋጥማችሁ ያውቃል?☐አዎ ☐አይ

a. መልስዎ አዎ ከሆነ ምን ችግር አጋጠምዎ?

3.______________________________

4.______________________________

3. ህጋዊ ያልሆነ መታወቅያ አጋጥምአቹ ያቃል?

☐አያውቅም ☐ያቃል፤ብዙ ጊዜ

☐ያቃል፤አንዳንን ጊዜ ☐ልያጋጥመኝ ይችላል አላውቅም፤የማውቅበት መንገድ የለም

4. በስራችሁ ምክንያት ደክሞአችሁ ወይም ተሰላችታችሁ ታቃላችሁ? በምን ያህል ?

☐አያውቅም ☐ያቃል፤ብዙ ጊዜ

☐ያቃል፤አንዳንን ጊዜ ☐ልያጋጥመኝ ይችላል አላውቅም፤የማውቅበት መንገድ የለም

5. እስከ ቅርብ ጊዜ አገልግሎት ላያ በነበረ የቀድሞው አሰራር ደስተኛ ነበራችሁ? የ የቀድሞው አሰራር ለናንተ

☐በጣም ጥሩ ነበር ☐ለኩፉ አይሰትም

☐ጥሩ ነበር ☐በጣም ያስጠላል ነበር

6. እንደ አገልጋይ(ሰራተኛ)ምን ምን አገልግሎት ለተቃሚ ትሰጣላችሁ?

1. ______________________________

2. ______________________________

7. በጊዜው ያሉ ችግሮችን ለመቅረፍ ኤሌክትሮኒክስ ወይም ኮምፕዩተራይዝድ የሆነ ስይስተም እየሰራን ነው፣ የናንተ ስራ በሚያቅል እና ለናንተ በሚመች መንገድ ስይስተሙ እንዲሰራ ምን ምን ነገሮችን ቢያካትት ጥሩ ነው ፤ ይመችናል ብላችሁ ታስባላችሁ(ከ ግልዎ እይታ፣ጥቅም እና ፍላጎት ኣንጻር)?
   1. ______________________________
   2. ______________________________

Questioner paper For Police officer

- ይሄ መጠይቅ ከ ፖሊስ ወይም ህጋዊ ኣካል መረጃ ለመሰብሰብ የተዘጋጀ ነው ፤ እና እባክዎ ያዩት፣የሚያውቁት፣የሰሙት እና ባጠቃላይ ያሎት መረጃ በሃላፊነት እንዲሞሉ በትህትና እንጠይቃለን፡፡
- ከታች ላሉትጢያቄዎች ሁለት ኣይነት መልስ ኣላቸው፡፡ ኣንደኛ ከተሰጡት ምርቻዎች ኣንዱ በማጥቆር መምረጥ ሲሆን ሁለተኛው በተሰጡት ክፍት ቦታዎች የምታቁት ሃሳብ መሙላት ነው ፡፡

1. የዜጎች መታወቅያ በስራ ሂደት ላይ በምን ኢጋጣሚ ወይም በምን ሁኔታ ወይም ለምን ነው የምትጠቀመው?
   1. ______________________________
   2. ______________________________
2. ከመታወቅያው ምን ምን መረጃ ነው ለናንተ የሚያስፈልግዎ ወይም ለስራህ ሚያስፈልግዎ ?
   1. ______________________________
   2. ______________________________
3. ኣብዛኛው ጊዜ ህጋዊ ባልሆነ መታወቂያ ወንጀል ይሰራል፣ የእርስዎ እይታስ ምንድነው?

☐የተሰራ ወንጀል አውቃለሁ ☐ወንጀል አኔን ኢጋጥሞኝ ያውቃል

☐የተሰራ ወንጀል በወሬ ደረጃ ሰምቸ አውቃለሁ ☐የተሰራ ወንጀል ኣላውቅም

   a. ምንድነው? እንዴት? ለመልስዎ ማብራሪያ ይስጡ
      1. ______________________________
      2. ______________________________

4. አንዳንድ ተገልጋዮች ህጋዊ ያልሆነ መታወቂያ ይዘው የሚገኙበት ሁኔታ ሊፈጠር ይችላል፡፡ የመታወቂያ ህጋዊነት ማረጋገጫ ስልቶች ኣላችሁ?

☐ኣዎ ☐ኣይ

   a. ስልቶች ካላችሁ ፤ ምን ምንድ ናቸው?
      3. ______________________________
      4. ______________________________
   ii. እነዚህ የምትጠቀሚቸው ስልቶች ወይም መንገዶች አስተማማኝ እና ውጤታማ ናቸው ብላችሁ ታስባላችሁ ?
      ☐ኣዎ ☐ኣይ

እንዴት? ለመልስአ ማብራሪያ ይስጡ፡

iii. ______________________________

iv. ______________________________

5. በጊዜው ያሉ ችግሮችን ለመቅረፍ ኤሌክትሮኒክስ ወይም ኮምፒዩተራይዝድ የሆነ ስይስተም እየሰራን ነው፣ የናንተ ስራ በሚያቅል እና ለናንተ በሚመች መንገድ ስይስተሙ እንዲሰራ ምን ምን ነገሮችን ቢያካትት ጥሩ ነው ፤ ይመችናል ብላችሁ ታስባላችሁ(ክ ግልዎ እይታ፣ጥቅም እና ፍላጎት አንጻር)?

   1. ______________________________